
Thresholded Local Hyper-Flow Diffusion

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Abstract

Local Hyper-Flow Diffusion (HFD) [Fountoulakis et al., 2021] gives an edge-size-independent Cheeger-type guarantee for seeded clustering in general submodular hypergraphs, but existing HFD solvers do not keep intermediate computation local at every iteration. We introduce THRESHOLDED LOCAL HFD (TL-HFD), a first-order method that maintains an active region around the seeds, performs projected subgradient updates on that region and its immediate boundary, and expands via thresholded (top- k) boundary activation. We prove that the local update is exact: the degree-preconditioned projected subgradient step restricted to the active region and its boundary coincides with the unrestricted global update. We establish finite-time dual suboptimality for both exact and thresholded updates, treating the latter as *inexact* projected subgradient steps with explicit skipped-boundary error. We further derive an additive activated-volume bound controlled by realized local subgradient norms and the minimum boundary-push among newly activated vertices, and translate approximate dual optimality with localized support into a robust sweep-cut guarantee for early-stopped iterates. For general submodular cut-costs, each iteration is local in the scanned region and oracle-sensitive in the hyperedge primitive. Empirically, TL-HFD often matches or improves over HFD while activating less volume, with the largest gains on noisy instances where diffusion tends to absorb non-target vertices.

1 Introduction

Many real-world systems involve higher-order interactions — such as co-purchasing users, co-expressed genes, and jointly-consumed items — for which hypergraphs provide a natural model by allowing each hyperedge to connect an arbitrary subset of vertices [Benson et al., 2016, Zhou et al., 2006, Agarwal et al., 2006]. A central problem is *local clustering*: given a small seed set, identify a nearby low-conductance cluster without computing a global partition. While diffusion-based methods such as approximate Pagerank provide a canonical solution in graphs [Andersen et al., 2006, Chung, 2007, Spielman and Teng, 2004], extending this paradigm to hypergraphs is more subtle because cutting a hyperedge is inherently ambiguous, with penalties ranging from cardinality-based costs to general submodular splitting functions [Lawler, 1973, Li and Milenkovic, 2018, Yoshida, 2019]. Local Hyper-Flow Diffusion (HFD) [Fountoulakis et al., 2021] addresses this by formulating seeded clustering in general submodular hypergraphs as a convex primal-dual program with an

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edge-size-independent Cheeger-type guarantee. Its primal also admits an alternating-minimization solver that is strongly local with respect to the solution vector, making HFD a compelling foundation for local hypergraph clustering under arbitrary submodular splitting costs.

This leaves a natural algorithmic question within the HFD framework: can diffusion be optimized by updates whose locality is explicit at every iteration, rather than only reflected in the sparsity of the final solution? HFD is typically optimized through a primal alternating-minimization procedure [Fountoulakis et al., 2021], which is effective in practice but does not guarantee that intermediate iterates remain local. Existing strongly-local hypergraph methods provide such locality under more restrictive splitting models, such as unit/cardinality cut-costs or bounded hyperedge size [Liu et al., 2021, Ibrahim and Gleich, 2020]. We therefore develop a complementary first-order method based on the HFD dual. Although the dual is non-smooth, its variables are precisely the diffusion heights used for sweep cuts, making it a natural object for local, early-stoppable optimization. Using this insight, we propose THRESHOLDED LOCAL HYPER-FLOW DIFFUSION (TL-HFD), a locality-by-design projected subgradient method that maintains a dynamically growing active region around the seeds and touches only this region and its immediate boundary. Boundary growth is controlled through thresholded top- k activations. Our analysis establishes finite-time convergence, locality, and sweep-cut guarantees for TL-HFD. For general submodular cut-costs, the per-iteration complexity is local in the scanned region and oracle-sensitive in the per-hyperedge evaluation model. In summary, our contributions are:

1. **A locality-by-design first-order optimizer for HFD.** We introduce TL-HFD, a first-order method for the non-smooth HFD dual that maintains a seed-anchored active region, updates only this region and its one-hop boundary, and expands through selective top- k boundary activation. Lemma 2 proves that this local degree-preconditioned projected subgradient update is exact: restricting the update to the active region and its boundary gives the same iterate as the unrestricted global projected subgradient step, since all vertices outside this local region are pushed negative and remain zero after projection.
2. **Finite-time optimization and sweep guarantees.** We prove finite-time dual suboptimality for TL-HFD. Theorem 1 gives the exact local PSGD rate in terms of restricted local subgradient norms, and Corollary 1 extends this to the thresholded top- k variant by treating skipped boundary vertices as an inexact projected subgradient error. We further show that approximate dual iterates with localized support admit a robust sweep guarantee (Theorem 2, Corollary 2).
3. **Activated-volume accounting.** We derive an additive activated-volume bound showing that the total volume promoted into the active region is controlled by realized local subgradient norms and the minimum boundary-push among newly activated vertices (Theorem 3). For general submodular cut-costs, the per-iteration complexity is local in the scanned region and oracle-sensitive in the underlying hyperedge primitive.
4. **Empirical evaluation.** Empirically, TL-HFD activates smaller volumes than HFD while improving sweep quality and F1 on noisier instances in both real-world and synthetic hypergraphs (Appendix D.10, D.11, D.12.2).

Related Work. We defer a detailed discussion of related work to Appendix A, where we survey hypergraph cut objectives, diffusion and Laplacian primitives, strongly-local clustering algorithms, and flow/convex formulations in context.

2 Preliminaries

Submodular hypergraphs. A weighted hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E}, \{(w_e, \vartheta_e)\}_{e \in \mathcal{E}})$ has vertex set $\mathcal{V}$ and hyperedges $e \subseteq \mathcal{V}$ with weights $\vartheta_e > 0$. Unlike ordinary edges, a hyperedge admits many splits of its vertices, each incurring a different penalty; these are encoded by a *cut-cost function* $w_e : 2^e \rightarrow \mathbb{R}_{\geq 0}$, assumed to be normalized ($w_e(\emptyset) = w_e(e) = 0$) and submodular. This framework subsumes unit, cardinality-based, and general submodular costs.

Degrees, volume, and conductance. The vertex degree is $d_v := \sum_{e \ni v} \vartheta_e$, with $D := \text{diag}(d)$. Throughout, we assume $d_v > 0$ for every $v \in \mathcal{V}$ (equivalently, $\mathcal{V}$ contains no isolated vertices); this is without loss of generality for seeded local clustering since isolated vertices are never reached by the diffusion. For $S \subseteq \mathcal{V}$, the *volume* is $\text{vol}(S) := \sum_{v \in S} d_v$ and the *boundary volume* is $\text{vol}(\partial S) := \sum_{e \in \mathcal{E}} \vartheta_e w_e(S \cap e)$, aggregating weighted penalties across the cut $(S, \mathcal{V} \setminus S)$. The *conductance* is

$\Phi(S) := \frac{\text{vol}(\partial S)}{\min\{\text{vol}(S), \text{vol}(\mathcal{V} \setminus S)\}}$. Low conductance indicates a set that is internally well-connected yet weakly coupled to the rest of the hypergraph. Direct minimization of Φ is intractable, motivating convex relaxations [Andersen et al., 2006, Fountoulakis et al., 2019, 2020, 2021, Liu et al., 2021].

Lovász extensions. To connect submodular cut-costs to convex optimization, Fountoulakis et al. [2021] use the Lovász extension of each w_e ,

$$f_e(x) := \max_{\rho \in B_e} \langle \rho, x \rangle, \quad B_e := \{\rho \in \mathbb{R}^{|e|} : \langle \rho, \mathbb{1}_S \rangle \leq w_e(S) \ \forall S \subseteq e, \ \langle \rho, \mathbb{1}_e \rangle = 0\},$$

with ρ extended by zero outside e . Since $f_e(\mathbb{1}_S) = w_e(S \cap e)$, the Lovász extension is the tightest convex extension of w_e to indicator vectors and provides a principled relaxation of discrete cut costs to continuous optimization.

Local hypergraph clustering: Given a submodular hypergraph $\mathcal{H}$ and a seed set $S \subset \mathcal{V}$, identify a set $\hat{C} \subset \mathcal{V}$ of low conductance $\Phi(\hat{C})$ that lies in the vicinity of S .

HFD diffusion objective. Following Fountoulakis et al. [2021], we adopt a convex relaxation that injects mass at the seeds and penalizes hyperedge splits via Lovász extensions. Given $\delta > 0$, define the seed injection $\Delta(v) := \delta d_v$ for $v \in S$ and $\Delta(v) := 0$ otherwise. The degree-weighted injection calibrates the seed signal to vertex connectivity. The quadratic term $\frac{\sigma}{2} x^\top D x$ makes F σ -strongly convex with respect to $\|\cdot\|_D$. The negative of the HFD dual objective is

$$\min_{x \in \mathbb{R}_+^{|\mathcal{V}|}} F(x) := \frac{1}{2} \sum_{e \in \mathcal{E}} \vartheta_e f_e(x)^2 + \frac{\sigma}{2} x^\top D x - \langle \Delta - d, x \rangle, \quad (1)$$

where $\sigma > 0$ controls degree-weighted regularization. Smaller σ lets the diffusion spread further; Equation (1) is the negative of the dual in HFD’s primal–dual flow formulation [Fountoulakis et al., 2021]; we write $\mathcal{D}(x) := -F(x)$ when connecting to sweep guarantees that build on HFD’s dual-based analysis. Our algorithm minimizes F directly. The solution $x^* \in \mathbb{R}_+^{|\mathcal{V}|}$ assigns nonnegative heights to vertices — the term $-\langle \Delta - d, x \rangle$ rewards height on sufficiently injected seeds and penalizes height on non-seeds, while $\frac{1}{2} \sum_e \vartheta_e f_e(x)^2$ penalizes within-hyperedge discrepancies and acts as an ℓ_2 -type penalty on induced cut costs across level sets. We extract a discrete cluster by a standard *sweep*: for thresholds h , form level sets $S_h := \{v \in \mathcal{V} : x_v^* \geq h\}$ and return the one with smallest conductance.

Notational convention. Throughout the paper, ∂ denotes either a vertex-set boundary, as in ∂S and $\text{vol}(\partial S)$, or the subdifferential of a convex function, as in $\partial F(x)$; the meaning is clear from context. We also use e and e^t for edges and error vectors, respectively, with context disambiguating the notation. For iterates $\{x^{(t)}\}_{t=0}^{T-1}$, we write $\hat{x} \in \arg \min_{0 \leq t \leq T-1} F(x^{(t)})$ to denote any best iterate, i.e., $\hat{x} = x^{(t^*)}$ for some $t^* \in \arg \min_{0 \leq t \leq T-1} F(x^{(t)})$ and use the analogous convention for $\arg \max$.

3 Thresholded Local Hyper-Flow Diffusion (TL-HFD)

Having formulated local hypergraph clustering, we now present Thresholded Local Hyper-Flow Diffusion (TL-HFD), a locality-by-design solver for the non-smooth objective (1).

Locality. For a vector $x \in \mathbb{R}_+^{|\mathcal{V}|}$, define its support $\text{supp}(x) = \{v \in \mathcal{V} : x_v > 0\}$ and the *seed-anchored active region* $A(x) := \text{supp}(x) \cup S$. The *boundary* of $A(x)$ is $\partial A(x) := \{u \notin A(x) : \exists e \in \mathcal{E}, u \in e \text{ and } e \cap A(x) \neq \emptyset\}$. The active region $A(x)$ contains all vertices currently participating in the diffusion; its boundary $\partial A(x)$ consists of vertices reachable in one hyperedge hop that have not yet been activated.

Local-set terminology. We distinguish three co-evolving vertex sets. The *active region* $A^{(t)} = \text{supp}(x^{(t)}) \cup S$ contains the vertices currently carrying mass. The *local computational region* $L_t := A^{(t)} \cup \partial A^{(t)}$ is the set touched at iteration t , including boundary vertices that are scored but not necessarily activated. Let $J_t \subseteq \partial A^{(t)}$ be the vertices first promoted into $A^{(t+1)}$ at iteration t ; the *activated set* $M_t := S \cup \bigcup_{s < t} J_s$ records every vertex ever activated. Top- k governs how quickly M_t grows, but does not reduce the cost of scoring the boundary, which remains proportional to $|\partial A^{(t)}|$ per iteration.

Nonsmoothness and subgradients. Each $f_e(x) = \max_{\rho \in B_e} \langle \rho, x \rangle$ from equation (1) is nondifferentiable in general; F is nonsmooth. By Danskin’s theorem [Danskin, 1967], for each hyperedge e any maximizer $\rho_e(x) \in M_e(x) := \arg \max_{\rho \in B_e} \langle \rho, x \rangle$ yields a subgradient of f_e at x (Appendix C.6). Selecting one such maximizer per hyperedge, define

$$g(x) := \sum_{e \in \mathcal{E}} \vartheta_e f_e(x) \rho_e(x) + \sigma D x - (\Delta - d). \quad (2)$$

Lemma 1. *Consider F defined in (1). For any x and any choice $\rho_e(x) \in M_e(x)$, the vector $g(x)$ in (2) satisfies $g(x) \in \partial F(x)$.*

Proof. See Appendix B.1. □

Degree-preconditioned PSGD. To solve equation (1), a standard projected subgradient step $x^{(t+1)} = \Pi_{\mathbb{R}_+^{|\mathcal{V}|}}(x^{(t)} - \eta_{t+1} g^{(t)})$ treats all vertices equally, but vertex degrees in hypergraphs can vary by orders of magnitude. To normalize the update across heterogeneous degrees, we apply the diagonal preconditioner D^{-1} : $x^{(t+1)} = \Pi_{\mathbb{R}_+^{|\mathcal{V}|}}(x^{(t)} - \eta_{t+1} D^{-1} g(x^{(t)}))$. (3)

Restricting to the active region. Degree preconditioning addresses scaling, but does not by itself avoid work on distant, irrelevant vertices for local clustering around the seed set. We maintain an explicit active region $A^{(t)} := A(x^{(t)})$ and its boundary $\partial A^{(t)} := \partial A(x^{(t)})$, and restrict all computation to $A^{(t)} \cup \partial A^{(t)}$. The following lemma shows that this restriction produces *exactly* the same iterates as the global update (3).

Lemma 2 (Locality of iterates). *Let $x \geq 0$ and $A := \text{supp}(x) \cup S$. For any subgradient selection $g(x)$ of the form (2), every vertex $u \notin A \cup \partial A$ satisfies $[g(x)]_u = d_u - \Delta(u) = d_u > 0$. Consequently, for every $\eta > 0$, $\left[\Pi_{\mathbb{R}_+^{|\mathcal{V}|}}(x - \eta D^{-1} g(x)) \right]_u = 0$. Thus, the global projected subgradient update can be implemented exactly by updating only $A^{(t)} \cup \partial A^{(t)}$ at iteration t .*

Proof. Full proof in Appendix B.2. □

Interpretation. Lemma 2 shows that locality is exact rather than heuristic: outside the active region and its one-hop boundary, the subgradient coordinate is positive, so the degree-preconditioned step moves the coordinate negative and projection returns it to zero. Hence the global update (3) can be implemented by the local update

$$x_v^{(t+1)} = \begin{cases} \left[x_v^{(t)} - \eta_{t+1} g_v(x^{(t)})/d_v \right]_+, & v \in A^{(t)} \cup \partial A^{(t)}, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Each coordinate $g_v(x^{(t)})$ depends only on hyperedges incident to v . Thus each iteration is local in the scanned region $L_t = A^{(t)} \cup \partial A^{(t)}$ and its incident hyperedges, rather than requiring an explicit pass over all of $\mathcal{V}$. We now pair the locality result with a standard finite-time analysis of projected subgradient descent, whose per-iteration cost scales with the scanned local region rather than explicitly with the full hypergraph.

Theorem 1 (ε -optimality of local PSGD). *Assume $d_v > 0$ for all $v \in \mathcal{V}$ and $\sigma > 0$. Let $x^* \in \arg \min_{x \geq 0} F(x)$. Run the local PSGD update $x^{(t+1)} = \Pi_{\mathbb{R}_+^{|\mathcal{V}|}}(x^{(t)} - \eta_{t+1} D^{-1} g^{(t)})$, $\eta_{t+1} = \frac{1}{\sigma(t+1)}$, where $g^{(t)} \in \partial F(x^{(t)})$ is selected as in equation (2). Let $L_t := A^{(t)} \cup \partial A^{(t)}$ and define the restricted norm $\|g^{(t)}\|_{D^{-1}, L_t}^2 := \sum_{v \in L_t} \frac{(g_v^{(t)})^2}{d_v}$. Let $\hat{x}^{(T)} \in \arg \min_{0 \leq t \leq T-1} F(x^{(t)})$. Then $F(\hat{x}^{(T)}) - F(x^*) \leq \frac{1}{2\sigma T} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1}$. In particular, if $G_{\text{loc}}^2(T) := \max_{0 \leq t < T} \|g^{(t)}\|_{D^{-1}, L_t}^2$, then $F(\hat{x}^{(T)}) - F(x^*) \leq \frac{G_{\text{loc}}^2(T)(1+\log T)}{2\sigma T}$.*

Proof. Full proof in Appendix B.3. □

Remark 1. The $(1+\log T)/T$ rate is standard for projected subgradient descent on σ -strongly convex nonsmooth objectives [Shamir and Zhang, 2013]; the contribution here is that (i) it applies to the HFD dual under degree-preconditioning and the specific subgradient selection of Lemma 1, (ii) by Lemma 2, each iteration is algorithmically local and touches only the local region $A^{(t)} \cup \partial A^{(t)}$, so the bound is achieved using only the scanned local region and its incident hyperedges, rather than requiring an explicit pass over the full hypergraph.

Remark 2 (Localized norm bound). Theorem 1 is stated in terms of the realized restricted local norms $\|g^{(t)}\|_{D^{-1}, L_t}$. Under the box-constraints of Proposition 1, $\|g^{(t)}\|_{D^{-1}, L_t}^2 \leq B^2 \text{vol}(L_t)$, $B = (\delta-1)(1/\sigma+2)$. Consequently, $G_{\text{loc}}^2(T) \leq B^2 \max_{0 \leq t < T} \text{vol}(L_t)$, and Theorem 1 yields $F(\hat{x}^{(T)}) - F(x^*) \leq \frac{B^2 \max_{0 \leq t < T} \text{vol}(L_t)(1+\log T)}{2\sigma T}$. Thus the convergence bound depends on the explored local regions, and not on $\text{vol}(\mathcal{V})$. The box projection is used only to obtain a uniform coordinatewise local subgradient bound; by Appendix C.2, it does not change the optimizer.

Activation bottleneck and per-iteration work. The local update (4) evaluates only hyperedges that intersect $L_t = A^{(t)} \cup \partial A^{(t)}$, so the per-iteration work is local in the scanned region, $O(\sum_{e: e \cap L_t \neq \emptyset} T_e)$, where T_e is the per-hyperedge cost (oracle-sensitive for general submodular cut-costs, depending on how f_e and $\rho_e(x^{(t)})$ are computed). Each iteration still inspects the current boundary $\partial A^{(t)}$. Boundary vertices with positive push $\kappa^{(t)}(u) := \max\{0, -g_u^{(t)}/d_u\}$ are candidates for promotion into the active region. Promoting all such vertices can rapidly enlarge $\text{vol}(L_t)$ and the support of the returned iterate. We therefore introduce thresholded top- k boundary activation. This controls the number of vertices promoted into $A^{(t+1)}$ but not the number inspected in $\partial A^{(t)}$. Accordingly, our locality claims have two parts: each iteration is local in the scanned region L_t , while the cumulative growth of the promoted set M_t is controlled separately by the additive activated-volume accounting bound in Theorem 3.

Thresholding the boundary. As noted above, the local update (4) must still inspect every boundary vertex in $\partial A^{(t)}$. Boundary vertices have a special structure: every boundary vertex satisfies $x_u^{(t)} = 0$ by construction, so its preconditioned projected update simplifies to a one-sided “push”: $x_u^{(t+1)} = \eta_{t+1} \kappa^{(t)}(u)$, where $\kappa^{(t)}(u) := \max\{0, -g_u^{(t)}/d_u\}$. The quantity $\kappa^{(t)}(u)$ measures the magnitude of the boundary push: larger values indicate a stronger first-order signal for increasing x_u . Algorithm 1 activates only the top- k boundary vertices by a score, setting $x_u^{(t+1)} = 0$ for the rest.

Committed boundary selection. At each iteration of Algorithm 1, TL-HFD activates the top- k boundary vertices ranked by the *score*, $s^{(t)}(u) = \kappa^{(t)}(u) \left(\frac{d_{\text{in}}^{(t)}(u)}{d_u} \right)^\gamma$, where $d_{\text{in}}^{(t)}(u)$ is the weighted indegree measuring *structural commitment* to the active set, and γ controls the balance between gradient signal and structural membership. For unit cut-cost, $d_{\text{in}}^{(t)}(u)$ counts edges incident to the active set; for cardinality and submodularity cut-costs, $d_{\text{in}}^{(t)}(u) = \sum_{e \ni u, e \cap A^{(t)} \neq \emptyset} \vartheta_e w_e(A^{(t)} \cap e)$ weights each edge by its overlap with the active set. Thresholding by selecting Top- k boundary vertices (ties are broken arbitrarily) for updates at each iteration introduces a truncation error supported only on skipped positively pushed boundary vertices, which we quantify in Lemma 3.

Lemma 3 (Thresholding as a truncated inexact step). *For each boundary vertex $u \in \partial A^{(t)}$, define $\kappa^{(t)}(u) := \max\{0, -\frac{g_u^{(t)}}{d_u}\}$. Let $K_t \subseteq \partial A^{(t)}$ be any subset of boundary vertices selected by Algorithm 1 for activation at iteration t . Let $x^{(t+1)}$ be the resulting thresholded iterate, and define the skipped positively pushed boundary set $B^{(t)} := \{u \in \partial A^{(t)} \setminus K_t : \kappa^{(t)}(u) > 0\}$. Define $e^{(t)} \in \mathbb{R}^{\mathcal{V}}$ by $e_u^{(t)} := \eta_{t+1} \kappa^{(t)}(u)$ for $u \in B^{(t)}$ and $e_u^{(t)} := 0$ otherwise. Then $x^{(t+1)} = \Pi_{\mathbb{R}^{\mathcal{V}}} (x^{(t)} - \eta_{t+1} D^{-1} g^{(t)} - e^{(t)})$. Moreover, $\|e^{(t)}\|_D^2 = \eta_{t+1}^2 \sum_{u \in B^{(t)}} d_u (\kappa^{(t)}(u))^2$, and $\|e^{(t)}\|_{1,D} := \sum_{u \in \mathcal{V}} d_u |e_u^{(t)}| = \eta_{t+1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u)$. If, in addition, the selected subgradient satisfies the coordinatewise local bound $\left| \frac{g_u^{(t)}}{d_u} \right| \leq B_{\text{loc}} \ \forall u \in A^{(t)} \cup \partial A^{(t)}$, then $\|e^{(t)}\|_D^2 \leq \eta_{t+1}^2 B_{\text{loc}}^2 \text{vol}(B^{(t)})$ and $\|e^{(t)}\|_{1,D} \leq \eta_{t+1} B_{\text{loc}} \text{vol}(B^{(t)})$ (under the hypotheses of Proposition 1, namely for the box-constrained variant with $0 \leq x_v^{(t)} \leq \bar{x} := \frac{\delta-1}{\sigma} \ \forall v$, one may take $B_{\text{loc}} = (\delta-1)(1/\sigma+2)$).*

Proof. Full proof in Appendix B.4. $\square$

Interpretation. Lemma 3 interprets thresholding as an inexact preconditioned PSGD step: its only error is the truncation vector $e^{(t)}$, supported on skipped positively pushed boundary vertices and controlled by $\sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u)$, or under Proposition 1, by $\text{vol}(B^{(t)})$ and B_{loc} . Thus boundary selection enters the convergence analysis as an explicit optimization penalty (Corollary 1). The bound is selection-rule agnostic: for any selected set, the penalty is determined by the skipped set $B^{(t)}$; for a fixed activation budget, it is minimized by retaining vertices with largest $d_u \kappa^{(t)}(u)$. Our implementation ranks by $s^{(t)}(u)$, which weights $\kappa^{(t)}(u)$ by a structural factor (γ) favoring vertices well attached to the active region (Algorithm 1); $\gamma > 0$ controls this bias (App. D.3).

Algorithm 1 Thresholded Local Hyper-Flow Diffusion (TL-HFD)

Require: Current iterate $x^{(t)}$, seed set S , step size η_{t+1} , activation count $k \geq 0$

Ensure: Next iterate $x^{(t+1)}$

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    # Active region and its one-hop boundary.
1:  $A^{(t)} \leftarrow \text{supp}(x^{(t)}) \cup S$ 
2:  $\partial A^{(t)} \leftarrow \{u \notin A^{(t)} : \exists e \in E, u \in e, e \cap A^{(t)} \neq \emptyset\}$ 
    # Compute subgradient only on the local region (Lemma 2).
3:  $g^{(t)} \leftarrow g(x^{(t)})$  restricted to  $A^{(t)} \cup \partial A^{(t)}$ 
4:  $x^{(t+1)} \leftarrow x^{(t)}$ 
    # Exact PSGD update on active vertices.
5: for all  $u \in A^{(t)}$  do
6:    $x_u^{(t+1)} \leftarrow \max\{0, x_u^{(t)} - \eta_{t+1} [g^{(t)}]_u / d_u\}$ 
7: end for
    # Scoring for boundary vertices.
8: for all  $u \in \partial A^{(t)}$  do
9:    $\kappa^{(t)}(u) \leftarrow \max\{0, -[g^{(t)}]_u / d_u\}$ 
10:   $c^{(t)}(u) \leftarrow (d_{\text{in}}^{(t)}(u) / d_u)^\gamma$ 
11:   $s^{(t)}(u) \leftarrow \kappa^{(t)}(u) c^{(t)}(u)$ 
12: end for
    # Activate the top-k (Thresholding).
13:  $K_t \leftarrow \text{TOPK}(s^{(t)}, k)$ 
14: for all  $u \in K_t$  do
15:   $x_u^{(t+1)} \leftarrow \eta_{t+1} \kappa^{(t)}(u)$ 
16: end for
17: return  $x^{(t+1)}$ 

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Corollary 1 (Suboptimality under boundary truncation). *Assume the hypotheses of Theorem 1. Run Algorithm 1 for T iterations with step sizes $\eta_{t+1} = 1/(\sigma(t+1))$. Let $\hat{x}^{(T)} \in \arg \min_{0 \leq t \leq T-1} F(x^{(t)})$ be a best iterate, and let $\bar{x} \geq \|x^*\|_\infty$. For each t , let $B^{(t)}$ be the skipped positively pushed boundary set from Lemma 3. Define $\varepsilon_{\text{loc}}(T) := \frac{1}{2\sigma T} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1}$. Then $F(\hat{x}^{(T)}) - F(x^*) \leq \varepsilon_{\text{loc}}(T) + \frac{\bar{x}}{T} \sum_{t=0}^{T-1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u)$. Moreover, if $|g_u^{(t)} / d_u| \leq B_{\text{loc}}$ for all $u \in A^{(t)} \cup \partial A^{(t)}$ and all $t < T$, then $F(\hat{x}^{(T)}) - F(x^*) \leq \varepsilon_{\text{loc}}(T) + \frac{\bar{x} B_{\text{loc}}}{T} \sum_{t=0}^{T-1} \text{vol}(B^{(t)}) =: \varepsilon_{\text{vol}}(T)$.*

Proof. Proof in Appendix B.5. $\square$

Remark 3 (Explicit constants). The explicit constants from Proposition 1 are formal for the box-constrained Algorithm 2 and transfer to Algorithm 1 whenever the box is non-binding (remark 7). If $G_{\text{loc}}^2(T) := \max_{0 \leq t < T} \|g^{(t)}\|_{D^{-1}, L_t}^2$, then Theorem 1 gives $\varepsilon_{\text{loc}}(T) \leq G_{\text{loc}}^2(T)(1 + \log T)/(2\sigma T)$. Under the hypotheses of Proposition 1 for the box-constrained variant, one may take $\bar{x} = (\delta - 1)/\sigma$ and $B_{\text{loc}} = (\delta - 1)(1/\sigma + 2)$. In that case, $\varepsilon_{\text{vol}}(T) = \varepsilon_{\text{loc}}(T) + \frac{(\delta-1)B_{\text{loc}}}{\sigma T} \sum_{t=0}^{T-1} \text{vol}(B^{(t)})$.

3.1 Sweep-Cut Guarantees

The convergence bounds above control the suboptimality gap, $F(\hat{x}^{(T)}) - F(x^*)$. To translate this into a clustering guarantee — i.e., a bound on the conductance of the sweep set extracted from $\hat{x}^{(T)}$ — we require structural assumptions relating the seed set to the target cluster, following Fountoulakis et al. [2021].

Assumption 1 (Seed overlap). There exists a target cluster $C \subset \mathcal{V}$ such that $\text{vol}(S \cap C) \geq \alpha \text{vol}(C)$ and $\text{vol}(S \cap C) \geq \beta \text{vol}(S)$ for some $\alpha, \beta \in (0, 1]$.²

Assumption 2 (Target volume). $\text{vol}(C) \leq \text{vol}(\mathcal{V} \setminus C)$. This is the standard assumption that the target cluster is the minority side of the cut, so that $\Phi(C) = \text{vol}(\partial C) / \text{vol}(C)$.

Assumption 3. Following Fountoulakis et al. [2021], we set $\delta = 3/\alpha$, assume $0 \leq w_e(A) \leq 1$ for all $e \in \mathcal{E}$, $A \subseteq e$, and require $\sigma \leq \beta\Phi(C)/3$.

The first condition requires that the seeds have sufficient coverage of the target cluster; the second requires that the seeds are sufficiently concentrated inside C . Throughout, we set the injection coefficient $\delta = 3/\alpha$, so that the seed mass is inversely proportional to the seed-cluster overlap; this choice is inherited from Fountoulakis et al. [2021].

Definition 1 ($\mathbb{K}$ -sweep locality). Fix a target cluster C and a vector $x \geq 0$. Define $\nu(x) := \sum_{e \in E} \vartheta_e f_e(x)^2$, and $\ell(e) := \begin{cases} f_e(x)/\sqrt{\nu(x)}, & \nu(x) > 0 \\ 0, & \nu(x) = 0 \end{cases}$. Let $\ell_b(e) := \max \left\{ \frac{1}{\sqrt{\text{vol}(C)}}, \ell(e) \right\}$.

We say that x satisfies $\mathbb{K}$ -sweep locality if $\sum_{e \in \mathcal{E}} \vartheta_e f_e(x) \ell_b(e) \leq \mathbb{K} \sqrt{\nu(x)}$.

The choice $\mathbb{K} = 4/\beta$ recovers the HFD sweep-locality condition $\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e) \leq \frac{4}{\beta} \sqrt{\nu(x)}$ [Fountoulakis et al., 2021, Claim A.2].

Theorem 2 (Sweep bound under $\mathbb{K}$ -sweep locality). Assume Assumptions 1, 2, 3 and let $x^* \in \arg \min_{x \geq 0} F(x)$. Let $x \geq 0$ satisfy the suboptimality $F(x) - F(x^*) \leq \varepsilon$. Assume that x satisfies $\mathbb{K}$ -sweep locality, $\sum_{e \in \mathcal{E}} \vartheta_e f_e(x) \ell_b(e) \leq \mathbb{K} \sqrt{\nu(x)}$. Define $D_C := \frac{\text{vol}(C)^2}{2 \text{vol}(\partial C)}$, and assume $\text{vol}(\partial C) > 0$. If $\varepsilon < D_C$, then there exists a threshold $h > 0$ such that the sweep set $S_h := \{v \in \mathcal{V} : x_v \geq h\}$ satisfies $\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3\mathbb{K}}{\alpha} \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1-\varepsilon/D_C}}$. If additionally $\text{vol}(S_h) \leq \text{vol}(\mathcal{V} \setminus S_h)$, then the same bound holds for conductance: $\Phi(S_h) \leq \frac{3\mathbb{K}}{\alpha} \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1-\varepsilon/D_C}}$.

Proof. Full proof in Appendix B.6. □

Theorem 2 applies to any approximate dual iterate satisfying $\mathbb{K}$ -sweep locality. To obtain an early-stopping guarantee for TL-HFD, we prove a support-volume sufficient condition in Appendix B.7: for any $x \geq 0$, x satisfies $\mathbb{K}$ -sweep locality with $\mathbb{K} = 1 + \sqrt{\frac{\text{vol}(\text{supp}(x))}{\text{vol}(C)}}$. Therefore, localized support of the returned iterate directly yields a sweep-locality constant.

Corollary 2 (Early-stopping sweep guarantee for TL-HFD). Assume Assumptions 1, 2, and 3. Assume also the hypotheses of Corollary 1, using the box-constrained Proposition 1 so that $\varepsilon_{\text{vol}}(T)$ is well-defined. Let $\hat{x}^{(T)} \in \arg \min_{0 \leq t < T} F(x^{(t)})$, and define $\zeta_T := \text{vol}(\text{supp}(\hat{x}^{(T)})) / \text{vol}(C)$. If $\varepsilon_{\text{vol}}(T) < D_C := \frac{\text{vol}(C)^2}{2 \text{vol}(\partial C)}$, then there exists a threshold $h > 0$ such that $S_h := \{v : \hat{x}_v^{(T)} \geq h\}$ satisfies $\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3}{\alpha} (1 + \sqrt{\zeta_T}) \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1-\varepsilon_{\text{vol}}(T)/D_C}}$. If additionally $\text{vol}(S_h) \leq \text{vol}(\mathcal{V} \setminus S_h)$, then the same bound holds for $\Phi(S_h)$.

Proof. Full proof in Appendix B.7. □

Remark 4 (Recovery of the HFD sweep-locality constant). If the returned iterate satisfies $\text{vol}(\text{supp}(\hat{x}^{(T)})) \leq \left(\frac{4}{\beta} - 1\right)^2 \text{vol}(C)$, then $1 + \sqrt{\zeta_T} \leq \frac{4}{\beta}$. In this case Corollary 2 recovers the same sweep constant as the HFD sweep-locality condition: $\frac{3}{\alpha} (1 + \sqrt{\zeta_T}) \leq \frac{12}{\alpha\beta}$.

²The experiments in Section 4 use seed injections of the form $\Delta[s^*] = \delta_{\text{exp}} \cdot \text{vol}(\mathcal{T})$ with $s^* \in C$ and $\mathcal{T} = C$. This corresponds to the degree-weighted form $\Delta(v) = \delta d_v$ for $v \in S = \{s^*\}$ with a seed-specific $\delta = \delta_{\text{exp}} \cdot \text{vol}(\mathcal{T}) / d_{s^*}$; see Proposition 3 in Appendix D.4 for further details.

Bound on cumulative activated volume. The bounds above are local in the scanned region $L_t = A^{(t)} \cup \partial A^{(t)}$, while the sweep guarantee depends on the support volume of the returned iterate. Top- k activation limits the number of promoted boundary vertices per iteration, but it does not by itself bound their degree volume or the cost of scoring the current boundary. We therefore track the set (M_t) of vertices ever promoted into the active region. The following theorem gives an additive trajectory-level bound on the activated volume $\text{vol}(M_T)$ in terms of restricted local subgradient energy and the realized boundary-push margins of newly activated vertices.

Theorem 3 (Explored-volume). *Assume $x^{(0)} = 0$. Let $M_0 := S$ and let $J_t := \{u \in \partial A^{(t)} : u \notin M_t, x_u^{(t+1)} > 0\}$ be the set of vertices activated for the first time at iteration t , where $M_{t+1} := M_t \cup J_t$. Equivalently, for boundary updates in Algorithm 1, $J_t = \{u \in K_t : u \notin M_t, \kappa^{(t)}(u) > 0\}$. Let $L_t := A^{(t)} \cup \partial A^{(t)}$ and $\kappa^{(t)}(u) := \max\left\{0, -\frac{g_u^{(t)}}{d_u}\right\}$. Let $I_T := \{0 \leq t < T : J_t \neq \emptyset\}$. Assume that for each $t \in I_T$, there exists $\rho_t > 0$ such that $\kappa^{(t)}(u) \geq \rho_t \forall u \in J_t$. Then, for $M_T = S \cup \bigcup_{t=0}^{T-1} J_t$, we have $\text{vol}(M_T) \leq \text{vol}(S) + \sum_{t \in I_T} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\rho_t^2}$. In particular, if $\|g^{(t)}\|_{D^{-1}, L_t} \leq G_{\text{loc}} \forall t$, then $\text{vol}(M_T) \leq \text{vol}(S) + G_{\text{loc}}^2 \sum_{t \in I_T} \rho_t^{-2}$.*

Proof. Full proof in Appendix C.4. $\square$

Remark 5 (Activated-volume accounting). Theorem 3 gives a trajectory-level accounting bound for the growth of the activated set. Along the TL-HFD trajectory, the volume promoted into M_T is controlled by two realized local quantities: the restricted subgradient energy $\|g^{(t)}\|_{D^{-1}, L_t}^2$ on the scanned region and the activation margin ρ_t of newly promoted boundary vertices. This separates two notions of locality: each iteration computes only on $L_t = A^{(t)} \cup \partial A^{(t)}$ and its incident hyperedges, while Theorem 3 tracks the cumulative growth of the state that is actually promoted. Since every nonzero coordinate of the returned best iterate is contained in M_T , this activated-volume bound provides the support-volume control used in the early-stopping sweep analysis.

From activated volume to sweep locality. Theorem 3 controls the activated set M_T . To use the support-volume sweep guarantee, we need to control the support volume of the returned best iterate $\hat{x}^{(T)}$. Since $\hat{x}^{(T)}$ is one of the iterates generated before time T , every nonzero coordinate of $\hat{x}^{(T)}$ is either a seed or a vertex activated at some earlier iteration. Hence $\text{supp}(\hat{x}^{(T)}) \subseteq M_T$. Therefore, under the hypotheses of Theorem 3, $\text{vol}(\text{supp}(\hat{x}^{(T)})) \leq \text{vol}(M_T) \leq \text{vol}(S) + \sum_{t \in I_T} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\rho_t^2}$, where ρ_t lower-bounds the positive boundary push of vertices first activated at iteration t . Combining this with the seed-concentration assumption $\text{vol}(S \cap C) \geq \beta \text{vol}(S)$ and the inclusion $S \cap C \subseteq C$, we have $\text{vol}(S) \leq \text{vol}(C)/\beta$. Thus $\frac{\text{vol}(\text{supp}(\hat{x}^{(T)}))}{\text{vol}(C)} \leq \frac{1}{\beta} + \frac{1}{\text{vol}(C)} \sum_{t \in I_T} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\rho_t^2}$. This gives an explicit support-volume bound that can be substituted into the early-stopping sweep guarantee through $\zeta_T = \text{vol}(\text{supp}(\hat{x}^{(T)}))/\text{vol}(C)$.

Summary of guarantees. The results above give an end-to-end local first-order interpretation of TL-HFD. Lemma 2 shows that each projected subgradient step can be implemented exactly on the scanned local region $L_t = A^{(t)} \cup \partial A^{(t)}$. Theorem 1 and Corollary 1 control the dual suboptimality of exact and thresholded updates. Theorem 2 and Corollary 2 translate approximate dual optimality and localized support into a sweep-cut guarantee for early-stopped iterates. Finally, Theorem 3 accounts for the cumulative volume promoted into the activated set M_T , which provides an explicit way to control the support ratio ζ_T appearing in the sweep bound. We next evaluate the performance of TL-HFD in practice.

4 Experiments

We evaluate TL-HFD (Algorithm 1 with top- k boundary scoring) on real-world and synthetic hypergraph datasets (Appendix D.10, D.11), comparing unit (U), cardinality (C), and submodular (S) cut-costs (Appendix D.2). On synthetic hypergraphs with varying target conductance, TL-HFD matches HFD on clean, well-separated clusters and improves over HFD on noisier clusters where unrestricted expansion tends to include non-target vertices (Appendix D.10). On real-world hypergraphs, TL-HFD gives the largest F1 gains on clusters where the baselines are weaker. Additional experiments on Amazon-reviews, Florida Bay, contact-high-school, oil trade, and synthetic HSBM are provided in Appendix D. The sensitivity analysis of γ from Algorithm 1 is in Appendix D.13.

Activation-scale selection. Following [Fountoulakis et al., 2021], we use target-volume-scaled seed injection: the injected mass is proportional to $\text{vol}(\mathcal{T})$, so the effective single-seed injection coefficient varies with the target cluster and seed degree. TL-HFD introduces one additional activation-scale parameter, the top- k fraction f . Rather than tuning the integer k directly, we set $k = \max(1, \lfloor f \cdot \text{vol}_{\text{est}} \rfloor)$ for unit cut-costs and $k = \max(1, \lfloor f \cdot n_{\text{est}} \rfloor)$ for cardinality cut-costs, where $\text{vol}_{\text{est}} = \max_v \Delta(v)/\delta_{\text{exp}}$ and $n_{\text{est}} = \text{vol}_{\text{est}}/\bar{d}$. We select f from a small grid and report the F1 score of the *lowest conductance cluster*. The sensitivity analysis of f is given in Appendix D.

Baseline Methods and parameters. We compare TL-HFD against HFD Alternating Minimization approach [Fountoulakis et al., 2021], LH- p [Liu et al., 2021] (localized quadratic hypergraph diffusions with $p \in \{1.4, 2.0\}$), and ACL [Andersen et al., 2006]. Complete details of all parameters used are provided in Appendix, D.3. We report median F1 score over all trials.

Trivago-Clicks. The Trivago-clicks dataset [Chodrow et al., 2021] is a hypergraph of hotel browsing sessions with $n = 172,738$ nodes, $m = 233,202$ hyperedges, and mean degree $\bar{d} \approx 5.6$. Following Fountoulakis et al. [2021], we evaluate on 10 destination-country clusters with $|\mathcal{T}| \in [144, 945]$. For each cluster, we run 100 single-seed trials with seeds selected uniformly at random from that cluster. We use injection factor $\delta_{\text{exp}} = 3$, set $\sigma = 10^{-4}$ and $\gamma = 1.0$, and run TL-HFD for 500 iterations for unit cut-costs and 1000 iterations for cardinality cut-costs. For each cluster we sweep f across the grid of fractions and select the configuration with the lowest median output conductance over the trials and report its median F1. Further hyperparameter details are provided in Appendix D.3.

Table 1: Trivago-clicks: median F1 score per cluster. Best result per cluster within each cut-cost block in **bold**. Cluster codes, and hyperparameters are reported in Appendix D.5.

Method	KOR	ISL	PRI	CRM	VNM	HKG	MLT	GTM	UKR	EST
U-HFD	0.73	0.96	0.85	0.82	0.22	0.95	0.97	0.88	0.58	0.93
TL-UHFD	0.86	0.98	0.97	0.85	0.58	0.65	0.97	0.95	0.65	0.79
U-LH-2.0	0.69	0.85	0.78	0.70	0.24	0.90	0.88	0.82	0.49	0.90
U-LH-1.4	0.69	0.84	0.79	0.75	0.28	0.87	0.92	0.83	0.47	0.90
ACL	0.64	0.84	0.75	0.68	0.23	0.90	0.83	0.69	0.50	0.88
C-HFD	0.75	0.98	0.93	0.93	0.29	0.80	0.97	0.97	0.68	0.53
TL-CHFD	0.82	0.98	0.97	0.85	0.32	0.76	0.97	0.97	0.73	0.58
C-LH-2.0	0.73	0.90	0.84	0.78	0.27	0.94	0.93	0.88	0.51	0.76
C-LH-1.4	0.71	0.88	0.84	0.78	0.27	0.88	0.91	0.85	0.50	0.78

Results. Table 1 summarizes the Trivago-clicks results. For unit cut-cost, TL-UHFD improves over U-HFD on 7/10 clusters and ties on one additional cluster. The largest gains occur on Vietnam, Puerto Rico, and South Korea, where the diffusion appears to benefit from a more selective boundary expansion. Intuitively, top- k scoring helps in these cases by preventing the diffusion from spreading through weakly related session hyperedges and instead prioritizing boundary vertices with both a large first-order push and stronger attachment to the current active region. The two clear losses, Hong Kong and Estonia, occur where U-HFD already obtains very high F1. These cases suggest a different failure mode: when the target cluster is already well recovered by the U-HFD, the top- k restriction can become too conservative or can delay activation of useful boundary vertices.

For cardinality cut-cost, TL-CHFD improves over C-HFD on 5/10 clusters, ties on 3/10, and loses on 2/10. Counting ties, TL-CHFD attains the best result within the cardinality block on 7/10 clusters. The gains are smaller than in the unit setting but more evenly distributed, consistent with the role of the cardinality weighting ($d_{\text{in}}(u) = \sum_{e \ni u, e \cap A \neq \emptyset} \min(|A \cap e|, |e \setminus A|)/\lfloor |e|/2 \rfloor$): rather than treating every nontrivial hyperedge split equally, it gives finer-grained credit to boundary vertices connected through hyperedges that overlap substantially with the active region.

Convergence and computational cost. HFD’s primal alternating-minimization solver converges in few iterations, while TL-HFD uses slower projected subgradient updates with the $O((1 + \log T)/T)$ rate of Theorem 1. The benefit of TL-HFD is that expansion is explicitly controlled: in single-seed runs, at most k new boundary vertices are promoted per iteration, so the maintained support grows much more gradually, although each iteration still scores the current boundary. Appendix D.12 studies this tradeoff on Amazon-reviews dataset. We observe that TL-HFD often reaches HFD-level F1 after only a small fraction of its 1000 iterations, while maintaining substantially smaller nonzero supports than HFD’s transient active sets.

Conclusion. We introduced THRESHOLDED LOCAL HYPER-FLOW DIFFUSION (TL-HFD), a locality-by-design first-order method for the non-smooth HFD dual on general submodular hypergraphs. TL-HFD maintains a seed-anchored active region, performs degree-preconditioned projected subgradient updates on this region and its one-hop boundary, and expands via thresholded top- k boundary activation. We provide an exact locality invariant, finite-time dual suboptimality for exact and thresholded updates, and a robust sweep guarantee for approximate iterates with localized support. We provide an activated-volume accounting bound in terms of realized local subgradient norms and boundary-push margins. Empirically, TL-HFD improves sweep quality and F1 on real-world and synthetic hypergraphs.

Limitations. Although top- k slows the growth of the active and activated sets, it does not remove the cost of scoring the current boundary. Thus the per-iteration work remains local in the scanned region L_t , rather than being controlled by top- k alone. Similarly, Theorem 3 is an activated-volume accounting bound: it controls the promoted volume in terms of realized local subgradient norms and boundary-push margins, but it is not an unconditional graph-size-independent runtime guarantee. A second limitation is optimization speed: projected subgradient descent has the finite-time rate $O((1 + \log T)/T)$, so it may require more iterations to reach high accuracy compared with HFD’s alternating minimization. Like HFD, TL-HFD uses a target-volume estimate; removing this dependence is left for future work.

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Appendix

A Related Work

Hypergraph cut objectives and hyperedge splitting models. A key challenge in hypergraph clustering is that cutting a hyperedge is inherently ambiguous: different applications induce different splitting penalties, which can be modeled through *cut-cost functions*. This viewpoint yields a useful hierarchy of models—from unit and cardinality-based penalties to general *submodular* costs that encode rich multiway structure [Lawler, 1973, Li and Milenkovic, 2018, Yoshida, 2019, Veldt et al., 2020, 2022]. Many practical approaches either restrict to simple penalties or reduce hypergraphs to graphs via clique/star expansions, which can introduce approximation effects that scale with hyperedge size and obscure higher-order structure.

Diffusion methods and Laplacian primitives for hypergraphs. Beyond cut-based formulations, numerous works extend random-walk, Pagerank, and heat-kernel ideas to hypergraphs by defining suitable Laplacians/transition rules and then applying diffusion-and-sweep pipelines [Chitra and Raphael, 2019, Hayashi et al., 2020, Takai et al., 2020, Chien et al., 2021]. Zhu and Segarra [2023] studied p -Laplacians for edge-dependent vertex weight (EDVW) hypergraphs by expressing EDVW splitting rules as submodular cut functions, enabling 1-Laplacian-based spectral clustering for richer higher-order structure. Recent work has developed efficient numerical primitives for *norm-based* (often non-smooth) hypergraph Laplacians, including algorithms for simulating heat diffusions and computing resolvents that recover Personalized Pagerank-type vectors and hypergraph Laplacian-system analogues [Ameranis et al., 2023]. Subsequent advances provide nearly-linear-time first-order algorithms for hypergraph Pagerank vectors and Laplacian systems over broad families of hypergraph potentials, including those induced by submodular cut functions [Ameranis et al., 2024]. However, these methods address a different class of hypergraph Laplacian objectives and are designed as global solvers. In contrast, we focus on locality-by-design optimization of the non-smooth dual arising in the HFD convex diffusion framework [Fountoulakis et al., 2021].

Local clustering and strongly-local algorithms. Local clustering in graphs is commonly studied through diffusion vectors and sweep cuts, with canonical guarantees linking personalized Pagerank-like quantities to low-conductance sets near a seed set [Andersen et al., 2006, Chung, 2007, Spielman and Teng, 2004]. A complementary line emphasizes *strong locality*, where runtime depends primarily on the explored/output volume rather than the full graph size, as in capacity-releasing diffusion [Wang et al., 2017]. Wei [2025] proposes alternative high-order conductance measures for hypergraphs and greedy seeded procedures aimed at real-time local clustering, offering a heuristic counterpart to diffusion/flow-based methods. In hypergraphs, scalable local clustering methods have been developed under additional structural restrictions (e.g., bounded hyperedge size) or for specialized objectives [Ibrahim and Gleich, 2020, Liu et al., 2021, Veldt et al., 2020]; Huang et al. [2024] study a different local primitive, namely localized densest subhypergraph discovery with negative supermodular objectives, and develop strongly local algorithms when the locality parameter exceeds a critical threshold. However, their framework targets density rather than conductance and is not aimed at optimizing the HFD dual. By contrast, we target the general submodular setting while designing updates that are local-by-design.

Flow/convex formulations and HFD-style local clustering. Flow-based and convex-optimization views unify diffusion dynamics with cut objectives in graphs [Arora et al., 2009, Fountoulakis et al., 2019, 2020, 2023]. For hypergraphs with expressive splitting models, *Local Hyper-Flow Diffusion (HFD)* [Fountoulakis et al., 2021] provides a convex primal–dual framework that supports general submodular cut-costs and yields *edge-size-independent* Cheeger-type guarantees for seeded clustering. Recent flow-based local hypergraph clustering variants include penalized flow objectives [Zhong et al., 2023] and approximation algorithms for generalized hypergraph ratio-cut objectives via max-flow reductions [Veldt, 2023]. In the global setting, Chen et al. [2025] generalize cut-matching games to hypergraphs with polymatroidal cut functions, yielding the first almost-linear-time $\mathcal{O}(\log n)$ approximation for partitioning general submodular hypergraphs. Their results address global balanced partitioning, rather than the seeded local clustering setting considered here. Our contribution is orthogonal to these formulations: we build directly on HFD’s dual

objective and develop a locality-by-design dual optimizer with finite-time suboptimality certificates that translate into robust sweep quality under early stopping.

Flow-based local hypergraph clustering beyond HFD. Several recent methods pursue flow-based seeded clustering objectives on hypergraphs that are adjacent to HFD but differ in formulation and optimization. For example, Penalized Flow Hypergraph Local Clustering introduces random-walk information as a penalization term within a flow-based framework and provides conductance-style guarantees [Zhong et al., 2023]. Related algorithmic work studies generalized hypergraph ratio-cut objectives via max-flow style reductions, e.g., using cut-matching games [Veldt, 2023].

Strongly-local hypergraph diffusions under restricted models. A complementary line develops strongly-local diffusion solvers for hypergraphs, typically for quadratic/cardinality-based cut objectives and/or under bounded hyperedge-size assumptions. Representative examples include strongly-local hypergraph quadratic diffusions [Liu et al., 2021] and capacity-releasing diffusion variants for hypergraphs [Ibrahim and Gleich, 2020]. More recent work explores alternative higher-order primitives (e.g., motif-based objectives) for local clustering in hypergraphs [Italiano et al., 2025].

Random-walk and Pagerank-style local clustering on hypergraphs. Beyond convex/flow formulations, Pagerank-style [Andersen et al., 2006] and random-walk-based [Chung, 2007, Gleich and Seshadhri, 2012, Kloster and Gleich, 2014] seeded clustering has been extended to more general graph models and applied to hypergraphs via EDVW-type representations, with proofs leveraging Lovász–Simonovits curve techniques [Li et al., 2024]. Related theoretical developments establish Lovász–Simonovits style results for hypergraph random walks [Kamal and Bagchi, 2024].

Spectral and variational learning on hypergraphs. A widely used approach to hypergraph learning constructs Laplacian-like operators (or relaxations) and applies spectral methods for clustering, classification, and embedding [Zhou et al., 2006, Hein et al., 2013]. This includes both practical algorithmic formulations and theoretical investigations of spectral partitioning in generative models [Ghoshdastidar and Dukkipati, 2014]. Variational formulations-including hypergraph total variation and related regularizers- offer an alternative route to learning with higher-order structure, often enabling robust semi-supervised learning and segmentation-style objectives [Hein et al., 2013].

B Theoretical Proofs

This appendix provides proofs for all formal results stated in the main body. Throughout, let $D = \text{diag}(d)$ with $d_u > 0$ for all $u \in V$, and define the weighted inner product and norms

$$\langle a, b \rangle_D := a^\top D b, \quad \|a\|_D := \sqrt{a^\top D a}, \quad \|g\|_{D^{-1}} := \sqrt{g^\top D^{-1} g}.$$

For a convex function F , we write $\partial F(x)$ for its subdifferential at x .

B.1 Proof of Lemma 1

Recall the subgradient selection in (2):

$$g(x) = \sum_{e \in E} \vartheta_e f_e(x) \rho_e(x) + \sigma D x - (\Delta - d), \quad \rho_e(x) \in M_e(x).$$

To show that $g(x) = \sum_{e \in E} \vartheta_e f_e(x) \rho_e(x) + \sigma D x - (\Delta - d) \in \partial F(x)$, we first show $f_e(x) \rho_e(x) \in \partial g_e(x)$.

Auxiliary Lemma 1 (Subgradient of the squared Lovász term). *Fix e and define $f_e(x) = \max_{\rho \in B_e} \langle \rho, x \rangle$ and $g_e(x) = \frac{1}{2} f_e(x)^2$. For any $x \in \mathbb{R}^V$ and any choice $\rho_e(x) \in M_e(x)$, the vector $f_e(x) \rho_e(x)$ is a subgradient of g_e at x :*

$$f_e(x) \rho_e(x) \in \partial g_e(x).$$

Proof. We use the following standard results.

Fact 1. For the support function $f_e(x) = \max_{\rho \in B_e} \langle \rho, x \rangle$ of a compact convex set B_e ,

$$\partial f_e(x) = \text{conv}(M_e(x)).$$

In particular, any maximizer $\rho_e(x) \in M_e(x)$ belongs to $\partial f_e(x)$.

Fact 2. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be convex, differentiable, and non-decreasing on the range of f , and let f be convex. Then

$$\partial(\phi \circ f)(x) = \phi'(f(x)) \cdot \partial f(x),$$

where the product denotes scalar multiplication of a set.

Here $\phi(s) = \frac{1}{2} s^2$ with $\phi'(s) = s$. Since the cut-cost is normalized, $0 \in B_e$, so $f_e(x) = \max_{\rho \in B_e} \langle \rho, x \rangle \geq \langle 0, x \rangle = 0$ for all x . Hence f_e takes values in $[0, \infty)$, where ϕ is non-decreasing, and Fact 2 applies. Therefore

$$\partial g_e(x) = \partial(\phi \circ f_e)(x) = \phi'(f_e(x)) \partial f_e(x) = f_e(x) \partial f_e(x).$$

Choosing $\rho_e(x) \in \partial f_e(x)$ gives $f_e(x) \rho_e(x) \in \partial g_e(x)$.

□

PROOF OF $g(x) \in \partial F(x)$:

Lemma 1. Consider F defined in (1). For any x and any choice $\rho_e(x) \in M_e(x)$, the vector $g(x)$ in (2) satisfies $g(x) \in \partial F(x)$.

Proof. Write

$$F(x) = \sum_{e \in E} \vartheta_e g_e(x) + \underbrace{\left(\frac{\sigma}{2} x^\top D x - \langle \Delta - d, x \rangle \right)}_{=: q(x)}.$$

By Auxiliary Lemma 1, for each e we have $\vartheta_e f_e(x) \rho_e(x) \in \partial(\vartheta_e g_e)(x)$. Since the subdifferential of a finite sum of convex functions equals the Minkowski sum of subdifferentials,

$$\sum_{e \in E} \vartheta_e f_e(x) \rho_e(x) \in \partial \left(\sum_{e \in E} \vartheta_e g_e \right) (x).$$

Moreover q is differentiable with $\nabla q(x) = \sigma Dx - (\Delta - d)$, and for a differentiable convex function $\partial q(x) = \{\nabla q(x)\}$. Therefore,

$$\partial F(x) = \partial \left(\sum_{e \in E} \vartheta_e g_e \right) (x) + \nabla q(x),$$

and selecting the element $\sum_e \vartheta_e f_e(x) \rho_e(x)$ from the first term gives

$$g(x) = \sum_{e \in E} \vartheta_e f_e(x) \rho_e(x) + \sigma Dx - (\Delta - d) \in \partial F(x).$$

□

B.2 Proof of Lemma 2: Locality of iterates

Without loss of generality, for the following proof we assume that there are no isolated vertices: as we compute D^{-1} (i.e., for any vertex u , $d_u > 0$).

Lemma 2 (Locality of iterates). *Let $x \geq 0$ and $A := \text{supp}(x) \cup S$. For any subgradient selection $g(x)$ of the form (2), every vertex $u \notin A \cup \partial A$ satisfies $[g(x)]_u = d_u - \Delta(u) = d_u > 0$. Consequently, for every $\eta > 0$, $\left[\Pi_{\mathbb{R}_+^{|V|}}(x - \eta D^{-1}g(x)) \right]_u = 0$. Thus, the global projected subgradient update can be implemented exactly by updating only $A^{(t)} \cup \partial A^{(t)}$ at iteration t .*

Proof. Fix an iteration t and suppose $\text{supp}(x^{(t)}) \subseteq A^{(t)}$, where $A^{(t)} := \text{supp}(x^{(t)}) \cup S$. Let $u \notin A^{(t)} \cup \partial A^{(t)}$. Then for every hyperedge $e \ni u$ we have $e \cap A^{(t)} = \emptyset$ by definition of the boundary. Since $x^{(t)}$ is zero outside $A^{(t)}$, it follows that $x^{(t)}|_e \equiv 0$ on each incident hyperedge. Because $0 \in B_e$ (normalized base polytopes contain the origin), we have $f_e(0) = 0$, hence $f_e(x^{(t)}) = 0$ for every hyperedge incident to u .

Consider the subgradient $g^{(t)} \in \partial F(x^{(t)})$ constructed in equation (2). At coordinate u , all hyperedge contributions vanish because they are multiplied by $f_e(x^{(t)}) = 0$. Therefore,

$$[g^{(t)}]_u = \sigma d_u x_u^{(t)} - (\Delta(u) - d_u).$$

Since $u \notin A^{(t)}$, we have $x_u^{(t)} = 0$. Also $u \notin S$ because $S \subseteq A^{(t)}$, hence $\Delta(u) = 0$, so $\Delta(u) - d_u = -d_u$. Thus $[g^{(t)}]_u = d_u > 0$.

The unprojected update at u is

$$y_u := x_u^{(t)} - \eta_{t+1} \frac{[g^{(t)}]_u}{d_u} = 0 - \eta_{t+1} \cdot 1 = -\eta_{t+1} < 0,$$

so after projection onto $\mathbb{R}_+^V$ we obtain $x_u^{(t+1)} = \max\{y_u, 0\} = 0$. Hence all coordinates outside $A^{(t)} \cup \partial A^{(t)}$ remain identically zero. □

B.3 Proof of Theorem 1: ϵ -optimality of local PSGD.

Auxiliary Lemma 2 (Strong convexity and subgradients). *Let F be σ -strongly convex w.r.t. $\|\cdot\|_D$. Then for every x, y and every $g \in \partial F(x)$,*

$$F(y) \geq F(x) + \langle g, y - x \rangle + \frac{\sigma}{2} \|y - x\|_D^2.$$

Equivalently,

$$\langle g, x - y \rangle \geq F(x) - F(y) + \frac{\sigma}{2} \|x - y\|_D^2.$$

Proof. This is a standard equivalent characterization of strong convexity: apply the definition of strong convexity using the supporting hyperplane at x given by a subgradient $g \in \partial F(x)$. $\square$

Theorem 1 (ϵ -optimality of local PSGD). *Assume $d_v > 0$ for all $v \in \mathcal{V}$ and $\sigma > 0$. Let $x^* \in \arg \min_{x \geq 0} F(x)$. Run the local PSGD update $x^{(t+1)} = \Pi_{\mathbb{R}_+^{|\mathcal{V}|}}(x^{(t)} - \eta_{t+1} D^{-1} g^{(t)})$, $\eta_{t+1} = \frac{1}{\sigma(t+1)}$, where $g^{(t)} \in \partial F(x^{(t)})$ is selected as in equation (2). Let $L_t := A^{(t)} \cup \partial A^{(t)}$ and define the restricted norm $\|g^{(t)}\|_{D^{-1}, L_t}^2 := \sum_{v \in L_t} \frac{(g_v^{(t)})^2}{d_v}$. Let $\hat{x}^{(T)} \in \arg \min_{0 \leq t \leq T-1} F(x^{(t)})$. Then $F(\hat{x}^{(T)}) - F(x^*) \leq \frac{1}{2\sigma T} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1}$. In particular, if $G_{\text{loc}}^2(T) := \max_{0 \leq t < T} \|g^{(t)}\|_{D^{-1}, L_t}^2$, then $F(\hat{x}^{(T)}) - F(x^*) \leq \frac{G_{\text{loc}}^2(T)(1+\log T)}{2\sigma T}$.*

Proof. Let

$$r^{(t)} := x^{(t)} - x^*.$$

Fix an iteration t , and write

$$L_t := A^{(t)} \cup \partial A^{(t)}, \quad F_t := \mathcal{V} \setminus L_t.$$

Define the unprojected point

$$y^{(t)} := x^{(t)} - \eta_{t+1} D^{-1} g^{(t)}.$$

By Lemma 2, for every $v \in F_t$, we have

$$x_v^{(t)} = 0, \quad g_v^{(t)} = d_v.$$

Therefore

$$y_v^{(t)} = x_v^{(t)} - \eta_{t+1} \frac{g_v^{(t)}}{d_v} = -\eta_{t+1} < 0.$$

Projection onto the nonnegative orthant gives

$$x_v^{(t+1)} = 0 \quad \forall v \in F_t.$$

Hence, for every $v \in F_t$,

$$r_v^{(t+1)} = x_v^{(t+1)} - x_v^* = -x_v^* = x_v^{(t)} - x_v^* = r_v^{(t)}.$$

Thus the contribution of F_t to the squared D -norm is identical at iterations t and $t+1$.

On the local region L_t , projection is nonexpansive coordinatewise in the D -norm. Therefore

$$\begin{aligned} \|r^{(t+1)}\|_D^2 &\leq \|r^{(t)}\|_D^2 - 2\eta_{t+1} \sum_{v \in L_t} g_v^{(t)} r_v^{(t)} + \eta_{t+1}^2 \sum_{v \in L_t} \frac{(g_v^{(t)})^2}{d_v} \\ &= \|r^{(t)}\|_D^2 - 2\eta_{t+1} \sum_{v \in L_t} g_v^{(t)} r_v^{(t)} + \eta_{t+1}^2 \|g^{(t)}\|_{D^{-1}, L_t}^2. \end{aligned}$$

We now compare the local inner product to the full inner product. Since $g_v^{(t)} = d_v$ and $r_v^{(t)} = -x_v^* \leq 0$ for $v \in F_t$,

$$\sum_{v \in F_t} g_v^{(t)} r_v^{(t)} = - \sum_{v \in F_t} d_v x_v^* \leq 0.$$

Consequently,

$$\sum_{v \in L_t} g_v^{(t)} r_v^{(t)} = \langle g^{(t)}, r^{(t)} \rangle - \sum_{v \in F_t} g_v^{(t)} r_v^{(t)} \geq \langle g^{(t)}, r^{(t)} \rangle.$$

Since F is σ -strongly convex with respect to $\|\cdot\|_D$,

$$\langle g^{(t)}, r^{(t)} \rangle \geq F(x^{(t)}) - F(x^*) + \frac{\sigma}{2} \|r^{(t)}\|_D^2.$$

Combining the preceding inequalities gives

$$\|r^{(t+1)}\|_D^2 \leq \|r^{(t)}\|_D^2 - 2\eta_{t+1} \left(F(x^{(t)}) - F(x^*) + \frac{\sigma}{2} \|r^{(t)}\|_D^2 \right) + \eta_{t+1}^2 \|g^{(t)}\|_{D^{-1}, L_t}^2.$$

Rearranging,

$$2\eta_{t+1} (F(x^{(t)}) - F(x^*)) \leq (1 - \sigma\eta_{t+1}) \|r^{(t)}\|_D^2 - \|r^{(t+1)}\|_D^2 + \eta_{t+1}^2 \|g^{(t)}\|_{D^{-1}, L_t}^2.$$

Using

$$\eta_{t+1} = \frac{1}{\sigma(t+1)}$$

gives

$$1 - \sigma\eta_{t+1} = \frac{t}{t+1}.$$

Multiplying by $t+1$, we obtain

$$\frac{2}{\sigma} (F(x^{(t)}) - F(x^*)) \leq t \|r^{(t)}\|_D^2 - (t+1) \|r^{(t+1)}\|_D^2 + \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\sigma^2(t+1)}.$$

Summing over $t = 0, \dots, T-1$, the norm terms telescope:

$$\sum_{t=0}^{T-1} \left(t \|r^{(t)}\|_D^2 - (t+1) \|r^{(t+1)}\|_D^2 \right) = -T \|r^{(T)}\|_D^2 \leq 0.$$

Therefore

$$\frac{2}{\sigma} \sum_{t=0}^{T-1} (F(x^{(t)}) - F(x^*)) \leq \frac{1}{\sigma^2} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1}.$$

Multiplying by $\sigma/2$,

$$\sum_{t=0}^{T-1} (F(x^{(t)}) - F(x^*)) \leq \frac{1}{2\sigma} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1}.$$

Since the best iterate is no worse than the average,

$$F(\hat{x}^{(T)}) - F(x^*) \leq \frac{1}{2\sigma T} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1}.$$

Finally, if

$$G_{\text{loc}}^2(T) = \max_{0 \leq t < T} \|g^{(t)}\|_{D^{-1}, L_t}^2,$$

then

$$\sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1} \leq G_{\text{loc}}^2(T) \sum_{t=0}^{T-1} \frac{1}{t+1} \leq G_{\text{loc}}^2(T) (1 + \log T).$$

Thus

$$F(\hat{x}^{(T)}) - F(x^*) \leq \frac{G_{\text{loc}}^2(T) (1 + \log T)}{2\sigma T}.$$

□

B.4 Proof of Lemma 3. Thresholding equals a truncated preconditioned step

The argument follows the standard inexact projected-gradient analysis Schmidt et al. [2011], specialized to the truncation error from Lemma 3.

Lemma 3 (Thresholding as a truncated inexact step). *For each boundary vertex $u \in \partial A^{(t)}$, define $\kappa^{(t)}(u) := \max\left\{0, -\frac{g_u^{(t)}}{d_u}\right\}$. Let $K_t \subseteq \partial A^{(t)}$ be any subset of boundary vertices selected by Algorithm 1 for activation at iteration t . Let $x^{(t+1)}$ be the resulting thresholded iterate, and define the skipped positively pushed boundary set $B^{(t)} := \{u \in \partial A^{(t)} \setminus K_t : \kappa^{(t)}(u) > 0\}$. Define $e^{(t)} \in \mathbb{R}^{|\mathcal{V}|}$ by $e_u^{(t)} := \eta_{t+1} \kappa^{(t)}(u)$ for $u \in B^{(t)}$ and $e_u^{(t)} := 0$ otherwise. Then $x^{(t+1)} = \Pi_{\mathbb{R}^{|\mathcal{V}|}}(x^{(t)} - \eta_{t+1} D^{-1} g^{(t)} - e^{(t)})$. Moreover, $\|e^{(t)}\|_D^2 = \eta_{t+1}^2 \sum_{u \in B^{(t)}} d_u (\kappa^{(t)}(u))^2$, and $\|e^{(t)}\|_{1,D} := \sum_{u \in \mathcal{V}} d_u |e_u^{(t)}| = \eta_{t+1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u)$. If, in addition, the selected subgradient satisfies the coordinatewise local bound $\left|\frac{g_u^{(t)}}{d_u}\right| \leq B_{\text{loc}} \quad \forall u \in A^{(t)} \cup \partial A^{(t)}$, then $\|e^{(t)}\|_D^2 \leq \eta_{t+1}^2 B_{\text{loc}}^2 \text{vol}(B^{(t)})$ and $\|e^{(t)}\|_{1,D} \leq \eta_{t+1} B_{\text{loc}} \text{vol}(B^{(t)})$ (under the hypotheses of Proposition 1, namely for the box-constrained variant with $0 \leq x_v^{(t)} \leq \bar{x} := \frac{\delta-1}{\sigma} \quad \forall v$, one may take $B_{\text{loc}} = (\delta-1)(1/\sigma+2)$).*

Proof. Let

$$y^{(t)} := x^{(t)} - \eta_{t+1} D^{-1} g^{(t)}.$$

By definition of $e^{(t)}$, the vector

$$\bar{y}^{(t)} := y^{(t)} - e^{(t)}$$

agrees with $y^{(t)}$ on all coordinates outside $B^{(t)}$, and satisfies

$$\bar{y}_u^{(t)} = 0 \quad \forall u \in B^{(t)}.$$

Indeed, if $u \in B^{(t)}$, then $u \in \partial A^{(t)}$, so $x_u^{(t)} = 0$. Moreover $\kappa^{(t)}(u) > 0$, hence $g_u^{(t)} < 0$, and therefore

$$y_u^{(t)} = -\eta_{t+1} \frac{g_u^{(t)}}{d_u} = \eta_{t+1} \kappa^{(t)}(u) = e_u^{(t)}.$$

Thus $\bar{y}_u^{(t)} = y_u^{(t)} - e_u^{(t)} = 0$ on $B^{(t)}$.

We now check the projected update coordinatewise.

First, if $u \in A^{(t)}$, then $u \notin B^{(t)}$, so $\bar{y}_u^{(t)} = y_u^{(t)}$. Algorithm 1 performs the exact projected PSGD update on active vertices, hence

$$x_u^{(t+1)} = \max\{0, y_u^{(t)}\} = \max\{0, \bar{y}_u^{(t)}\}.$$

Second, if $u \in K_t$, then $u \notin B^{(t)}$, so again $\bar{y}_u^{(t)} = y_u^{(t)}$. Since $u \in \partial A^{(t)}$, we have $x_u^{(t)} = 0$. Therefore

$$\max\{0, y_u^{(t)}\} = \eta_{t+1} \kappa^{(t)}(u),$$

which is exactly the boundary activation performed by the algorithm.

Third, if $u \in \partial A^{(t)} \setminus K_t$ and $\kappa^{(t)}(u) = 0$, then $g_u^{(t)} \geq 0$. Since $x_u^{(t)} = 0$,

$$y_u^{(t)} = -\eta_{t+1} \frac{g_u^{(t)}}{d_u} \leq 0.$$

The algorithm leaves this coordinate at zero, and

$$\max\{0, \bar{y}_u^{(t)}\} = \max\{0, y_u^{(t)}\} = 0.$$

Fourth, if $u \in B^{(t)}$, then by construction

$$\bar{y}_u^{(t)} = 0.$$

The algorithm skips this positively pushed boundary vertex and leaves $x_u^{(t+1)} = 0$. Hence

$$x_u^{(t+1)} = \max\{0, \bar{y}_u^{(t)}\}.$$

Finally, if $u \notin A^{(t)} \cup \partial A^{(t)}$, then Lemma 2 gives

$$g_u^{(t)} = d_u > 0, \quad x_u^{(t)} = 0.$$

Hence

$$y_u^{(t)} = -\eta_{t+1} < 0.$$

The algorithm leaves this coordinate at zero, and

$$\max\{0, \bar{y}_u^{(t)}\} = \max\{0, y_u^{(t)}\} = 0.$$

Combining the cases,

$$x^{(t+1)} = \Pi_{\mathbb{R}_+^{|V|}}(\bar{y}^{(t)}) = \Pi_{\mathbb{R}_+^{|V|}}\left(x^{(t)} - \eta_{t+1} D^{-1} g^{(t)} - e^{(t)}\right).$$

The norm identities follow directly from the definition of $e^{(t)}$:

$$\|e^{(t)}\|_D^2 = \sum_{u \in V} d_u (e_u^{(t)})^2 = \eta_{t+1}^2 \sum_{u \in B^{(t)}} d_u (\kappa^{(t)}(u))^2,$$

and

$$\|e^{(t)}\|_{1,D} = \sum_{u \in V} d_u |e_u^{(t)}| = \eta_{t+1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u).$$

If the coordinatewise bound

$$|g_u^{(t)}|/d_u \leq B_{\text{loc}}$$

holds on $A^{(t)} \cup \partial A^{(t)}$, then for every $u \in B^{(t)} \subseteq \partial A^{(t)}$,

$$\kappa^{(t)}(u) = -\frac{g_u^{(t)}}{d_u} \leq \left| \frac{g_u^{(t)}}{d_u} \right| \leq B_{\text{loc}}.$$

Therefore

$$\|e^{(t)}\|_D^2 \leq \eta_{t+1}^2 B_{\text{loc}}^2 \sum_{u \in B^{(t)}} d_u = \eta_{t+1}^2 B_{\text{loc}}^2 \text{vol}(B^{(t)}),$$

and

$$\|e^{(t)}\|_{1,D} \leq \eta_{t+1} B_{\text{loc}} \sum_{u \in B^{(t)}} d_u = \eta_{t+1} B_{\text{loc}} \text{vol}(B^{(t)}).$$

The final statement follows from Proposition 1. □

B.5 Proof of Corollary 1: Suboptimality under boundary truncation

Corollary 1 (Suboptimality under boundary truncation). *Assume the hypotheses of Theorem 1. Run Algorithm 1 for T iterations with step sizes $\eta_{t+1} = 1/(\sigma(t+1))$. Let $\hat{x}^{(T)} \in \arg \min_{0 \leq t \leq T-1} F(x^{(t)})$ be a best iterate, and let $\bar{x} \geq \|x^*\|_\infty$. For each t , let $B^{(t)}$ be the skipped positively pushed boundary set from Lemma 3. Define $\varepsilon_{\text{loc}}(T) := \frac{1}{2\sigma T} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1}$. Then $F(\hat{x}^{(T)}) - F(x^*) \leq \varepsilon_{\text{loc}}(T) + \frac{\bar{x}}{T} \sum_{t=0}^{T-1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u)$. Moreover, if $|g_u^{(t)}/d_u| \leq B_{\text{loc}}$ for all $u \in A^{(t)} \cup \partial A^{(t)}$ and all $t < T$, then $F(\hat{x}^{(T)}) - F(x^*) \leq \varepsilon_{\text{loc}}(T) + \frac{\bar{x} B_{\text{loc}}}{T} \sum_{t=0}^{T-1} \text{vol}(B^{(t)}) =: \varepsilon_{\text{vol}}(T)$.*

Proof. Algorithm and box-constraint scope. The bound, involving the accumulated truncation error $\sum_{t < T} \|e^{(t)}\|_{1,D}$, uses only the hypotheses of Theorem 1, the thresholding-error characterization of Lemma 3, and the coordinate-wise bound $\|x^*\|_\infty \leq \bar{x}$ on the optimizer (Auxiliary Lemma 5). It applies to the unconstrained Algorithm 1 because $\Pi_{\mathbb{R}^{|V|}}$ is coordinatewise and 1-Lipschitz in the D -norm. The same argument applies to the box-constrained Algorithm 2 because $\Pi_{\mathcal{X}}$ is also coordinatewise and 1-Lipschitz in the D -norm (Auxiliary Lemma 6); for skipped boundary coordinates, the box-projected truncation error is $\min\{\bar{x}, \eta_{t+1} \kappa^{(t)}(u)\}$ and is therefore bounded by $\eta_{t+1} \kappa^{(t)}(u)$. The $\varepsilon_{\text{vol}}(T)$ specialization additionally invokes the uniform bound $|g_u^{(t)}/d_u| \leq B_{\text{loc}}$ from Proposition 1, which holds for Algorithm 2; for runs of Algorithm 1 whose iterates stay below $\bar{x}$, the two algorithms coincide (Remark 7).

The proof follows the localized inexact projected-subgradient argument. Define the exact projected local PSGD iterate

$$\tilde{x}^{(t+1)} := \Pi_{\mathbb{R}_+^{|V|}} \left(x^{(t)} - \eta_{t+1} D^{-1} g^{(t)} \right),$$

and define

$$d^{(t)} := x^{(t)} - x^*, \quad \tilde{d}^{(t+1)} := \tilde{x}^{(t+1)} - x^*, \quad d^{(t+1)} := x^{(t+1)} - x^*.$$

From the localized one-step inequality used in the proof of Theorem 1, we have

$$\begin{aligned} 2\eta_{t+1} (F(x^{(t)}) - F(x^*)) &\leq (1 - \sigma\eta_{t+1}) \|d^{(t)}\|_D^2 - \|\tilde{d}^{(t+1)}\|_D^2 \\ &\quad + \eta_{t+1}^2 \|g^{(t)}\|_{D^{-1}, L_t}^2. \end{aligned}$$

This is the exact-update progress inequality, with the global subgradient norm replaced by the restricted local norm on L_t .

We now compare the exact iterate $\tilde{x}^{(t+1)}$ with the thresholded iterate $x^{(t+1)}$. By Lemma 3, the two iterates can differ only on the skipped positively pushed set $B^{(t)}$. For $u \in B^{(t)}$, the exact update gives

$$\tilde{x}_u^{(t+1)} = e_u^{(t)} = \eta_{t+1} \kappa^{(t)}(u),$$

while the thresholded update leaves

$$x_u^{(t+1)} = 0.$$

Therefore

$$\begin{aligned} \|d^{(t+1)}\|_D^2 - \|\tilde{d}^{(t+1)}\|_D^2 &= \sum_{u \in B^{(t)}} d_u \left[(x_u^{(t+1)} - x_u^*)^2 - (\tilde{x}_u^{(t+1)} - x_u^*)^2 \right] \\ &= \sum_{u \in B^{(t)}} d_u \left[(x_u^*)^2 - (e_u^{(t)} - x_u^*)^2 \right]. \end{aligned}$$

Expanding the square,

$$(x_u^*)^2 - (e_u^{(t)} - x_u^*)^2 = 2e_u^{(t)} x_u^* - (e_u^{(t)})^2 \leq 2\bar{x} e_u^{(t)},$$

because $x^* \geq 0$, $e_u^{(t)} \geq 0$, and $\|x^*\|_\infty \leq \bar{x}$. Hence

$$\|d^{(t+1)}\|_D^2 \leq \|\tilde{d}^{(t+1)}\|_D^2 + 2\bar{x} \|e^{(t)}\|_{1,D}.$$

Substituting this into the exact-update one-step inequality gives

$$2\eta_{t+1}(F(x^{(t)}) - F(x^*)) \leq (1 - \sigma\eta_{t+1})\|d^{(t)}\|_D^2 - \|d^{(t+1)}\|_D^2 + \eta_{t+1}^2\|g^{(t)}\|_{D^{-1}, L_t}^2 + 2\bar{x}\|e^{(t)}\|_{1,D}.$$

Now use

$$\eta_{t+1} = \frac{1}{\sigma(t+1)}.$$

Then

$$1 - \sigma\eta_{t+1} = \frac{t}{t+1}.$$

Multiplying the preceding inequality by $t+1$, we obtain

$$\begin{aligned} \frac{2}{\sigma}(F(x^{(t)}) - F(x^*)) &\leq t\|d^{(t)}\|_D^2 - (t+1)\|d^{(t+1)}\|_D^2 \\ &\quad + \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\sigma^2(t+1)} + 2\bar{x}(t+1)\|e^{(t)}\|_{1,D}. \end{aligned}$$

Summing over $t = 0, \dots, T-1$, the first two terms telescope:

$$\sum_{t=0}^{T-1} [t\|d^{(t)}\|_D^2 - (t+1)\|d^{(t+1)}\|_D^2] = -T\|d^{(T)}\|_D^2 \leq 0.$$

Therefore

$$\begin{aligned} \frac{2}{\sigma} \sum_{t=0}^{T-1} (F(x^{(t)}) - F(x^*)) &\leq \frac{1}{\sigma^2} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1} \\ &\quad + 2\bar{x} \sum_{t=0}^{T-1} (t+1)\|e^{(t)}\|_{1,D}. \end{aligned}$$

Multiplying by $\sigma/2$,

$$\begin{aligned} \sum_{t=0}^{T-1} (F(x^{(t)}) - F(x^*)) &\leq \frac{1}{2\sigma} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1} \\ &\quad + \sigma\bar{x} \sum_{t=0}^{T-1} (t+1)\|e^{(t)}\|_{1,D}. \end{aligned}$$

By Lemma 3,

$$\|e^{(t)}\|_{1,D} = \eta_{t+1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u).$$

Since

$$(t+1)\eta_{t+1} = \frac{1}{\sigma},$$

we get

$$\sigma(t+1)\|e^{(t)}\|_{1,D} = \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u).$$

Hence

$$\sigma\bar{x} \sum_{t=0}^{T-1} (t+1)\|e^{(t)}\|_{1,D} = \bar{x} \sum_{t=0}^{T-1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u).$$

Therefore

$$\begin{aligned} \sum_{t=0}^{T-1} (F(x^{(t)}) - F(x^*)) &\leq \frac{1}{2\sigma} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1} \\ &\quad + \bar{x} \sum_{t=0}^{T-1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u). \end{aligned}$$

Dividing by T and using that the best iterate is no worse than the average,

$$\begin{aligned}
F(\hat{x}^{(T)}) - F(x^*) &\leq \frac{1}{2\sigma T} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1} \\
&\quad + \frac{\bar{x}}{T} \sum_{t=0}^{T-1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u) \\
&= \varepsilon_{\text{loc}}(T) + \frac{\bar{x}}{T} \sum_{t=0}^{T-1} \sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u).
\end{aligned}$$

This proves the bound.

If the coordinatewise local bound

$$|g_u^{(t)}|/d_u \leq B_{\text{loc}}$$

holds on $A^{(t)} \cup \partial A^{(t)}$, then for every $u \in B^{(t)}$,

$$\kappa^{(t)}(u) \leq B_{\text{loc}}.$$

Therefore

$$\sum_{u \in B^{(t)}} d_u \kappa^{(t)}(u) \leq B_{\text{loc}} \sum_{u \in B^{(t)}} d_u = B_{\text{loc}} \text{vol}(B^{(t)}).$$

Substituting gives

$$F(\hat{x}^{(T)}) - F(x^*) \leq \varepsilon_{\text{loc}}(T) + \frac{\bar{x} B_{\text{loc}}}{T} \sum_{t=0}^{T-1} \text{vol}(B^{(t)}) = \varepsilon_{\text{vol}}(T).$$

Finally, by Theorem 1,

$$\varepsilon_{\text{loc}}(T) = \frac{1}{2\sigma T} \sum_{t=0}^{T-1} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{t+1} \leq \frac{G_{\text{loc}}^2(T)(1 + \log T)}{2\sigma T},$$

where

$$G_{\text{loc}}^2(T) = \max_{0 \leq t < T} \|g^{(t)}\|_{D^{-1}, L_t}^2.$$

Under the hypotheses of Proposition 1 for the box-constrained variant, one may take

$$\bar{x} = \frac{\delta - 1}{\sigma} \quad \text{and} \quad B_{\text{loc}} = (\delta - 1)(1/\sigma + 2),$$

to get $\varepsilon_{\text{vol}}(T) = \varepsilon_{\text{loc}}(T) + \frac{(\delta-1)B_{\text{loc}}}{\sigma T} \sum_{t=0}^{T-1} \text{vol}(B^{(t)})$. □

B.6 Proof of Theorem 2: Sweep Bound under Dual Suboptimality

Auxiliary Lemma 3 (HFD lower bound on the optimal dual value). *Assume Assumptions 1, 2, and 3. In particular, assume $\delta = \frac{3}{\alpha}$, $0 \leq w_e(A) \leq 1 \quad \forall e \in \mathcal{E}$, $A \subseteq e$, and $\sigma \leq \frac{\beta \Phi(C)}{3}$. Let $D(x) := -F(x)$, $x^* \in \arg \min_{x \geq 0} F(x)$. Equivalently, $x^* \in \arg \max_{x \geq 0} D(x)$. Define $D_C := \frac{\text{vol}(C)^2}{2 \text{vol}(\partial C)}$, and assume $\text{vol}(\partial C) > 0$. Then $D(x^*) \geq D_C$.*

Proof. Since $x^* \in \arg \max_{x \geq 0} D(x)$, it suffices to exhibit one feasible vector $x \geq 0$ such that $D(x) \geq D_C$. We use the test vector

$$x = \lambda \mathbf{1}_C, \quad \lambda \geq 0.$$

Recall that

$$D(x) = \langle \Delta - d, x \rangle - \frac{1}{2} \sum_{e \in E} \vartheta_e f_e(x)^2 - \frac{\sigma}{2} x^\top D x.$$

By positive homogeneity of the Lovász extension,

$$f_e(\lambda \mathbf{1}_C) = \lambda f_e(\mathbf{1}_C).$$

Moreover,

$$f_e(\mathbf{1}_C) = w_e(C \cap e).$$

We first lower-bound the linear term. Since

$$\Delta(v) = \delta d_v \quad \text{for } v \in S$$

and $\Delta(v) = 0$ otherwise,

$$\begin{aligned} \langle \Delta - d, \mathbf{1}_C \rangle &= \Delta(C) - \text{vol}(C) \\ &= \delta \text{vol}(S \cap C) - \text{vol}(C). \end{aligned}$$

By Assumption 1,

$$\text{vol}(S \cap C) \geq \alpha \text{vol}(C).$$

Using $\delta = 3/\alpha$, we get

$$\langle \Delta - d, \mathbf{1}_C \rangle \geq \frac{3}{\alpha} \alpha \text{vol}(C) - \text{vol}(C) = 2 \text{vol}(C).$$

Therefore

$$\langle \Delta - d, \lambda \mathbf{1}_C \rangle \geq 2\lambda \text{vol}(C).$$

Next, since $0 \leq w_e(A) \leq 1$, we have

$$w_e(C \cap e)^2 \leq w_e(C \cap e).$$

Thus

$$\begin{aligned} \sum_{e \in E} \vartheta_e f_e(\mathbf{1}_C)^2 &= \sum_{e \in E} \vartheta_e w_e(C \cap e)^2 \\ &\leq \sum_{e \in E} \vartheta_e w_e(C \cap e) \\ &= \text{vol}(\partial C). \end{aligned}$$

Also,

$$\mathbf{1}_C^\top D \mathbf{1}_C = \text{vol}(C).$$

Combining these estimates gives

$$D(\lambda \mathbf{1}_C) \geq 2\lambda \text{vol}(C) - \frac{\lambda^2}{2} \text{vol}(\partial C) - \frac{\sigma \lambda^2}{2} \text{vol}(C).$$

Choose

$$\lambda = \frac{\text{vol}(C)}{\text{vol}(\partial C)}.$$

Then

$$\begin{aligned} D(\lambda \mathbf{1}_C) &\geq 2 \frac{\text{vol}(C)^2}{\text{vol}(\partial C)} - \frac{1}{2} \frac{\text{vol}(C)^2}{\text{vol}(\partial C)} - \frac{\sigma}{2} \frac{\text{vol}(C)^3}{\text{vol}(\partial C)^2} \\ &= \left(\frac{3}{2} - \frac{\sigma}{2} \frac{\text{vol}(C)}{\text{vol}(\partial C)} \right) \frac{\text{vol}(C)^2}{\text{vol}(\partial C)}. \end{aligned}$$

By Assumption 2,

$$\text{vol}(C) \leq \text{vol}(V \setminus C),$$

so

$$\Phi(C) = \frac{\text{vol}(\partial C)}{\text{vol}(C)}.$$

Since $\beta \leq 1$ and

$$\sigma \leq \frac{\beta \Phi(C)}{3},$$

we have

$$\sigma \leq \frac{\Phi(C)}{3} = \frac{\text{vol}(\partial C)}{3 \text{vol}(C)}.$$

Therefore

$$\sigma \frac{\text{vol}(C)}{\text{vol}(\partial C)} \leq \frac{1}{3}.$$

Substituting this into the previous lower bound yields

$$\begin{aligned} D(\lambda \mathbf{1}_C) &\geq \left(\frac{3}{2} - \frac{1}{6} \right) \frac{\text{vol}(C)^2}{\text{vol}(\partial C)} \\ &= \frac{4}{3} \frac{\text{vol}(C)^2}{\text{vol}(\partial C)}. \end{aligned}$$

In particular,

$$D(\lambda \mathbf{1}_C) \geq \frac{\text{vol}(C)^2}{2 \text{vol}(\partial C)} = D_C.$$

Since x^* maximizes D over $x \geq 0$, we conclude

$$D(x^*) \geq D(\lambda \mathbf{1}_C) \geq D_C.$$

□

Theorem 2 (Sweep bound under $\mathbb{K}$ -sweep locality). *Assume Assumptions 1, 2, 3 and let $x^* \in \arg \min_{x \geq 0} F(x)$. Let $x \geq 0$ satisfy the suboptimality $F(x) - F(x^*) \leq \varepsilon$. Assume that x satisfies $\mathbb{K}$ -sweep locality, $\sum_{e \in \mathcal{E}} \vartheta_e f_e(x) \ell_b(e) \leq \mathbb{K} \sqrt{\nu(x)}$. Define $D_C := \frac{\text{vol}(C)^2}{2 \text{vol}(\partial C)}$, and assume $\text{vol}(\partial C) > 0$. If $\varepsilon < D_C$, then there exists a threshold $h > 0$ such that the sweep set $S_h := \{v \in \mathcal{V} : x_v \geq h\}$ satisfies $\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3\mathbb{K}}{\alpha} \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1-\varepsilon/D_C}}$. If additionally $\text{vol}(S_h) \leq \text{vol}(\mathcal{V} \setminus S_h)$, then the same bound holds for conductance: $\Phi(S_h) \leq \frac{3\mathbb{K}}{\alpha} \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1-\varepsilon/D_C}}$.*

Proof. Let

$$D(x) := -F(x)$$

denote the HFD dual objective. Since

$$F(x) - F(x^*) \leq \varepsilon,$$

we equivalently have

$$D(x^*) - D(x) \leq \varepsilon.$$

By the auxiliary Lemma 3,

$$D(x^*) \geq D_C.$$

Therefore

$$D(x) \geq D(x^*) - \varepsilon \geq D_C - \varepsilon.$$

Since $\varepsilon < D_C$, this implies

$$D(x) > 0.$$

Define

$$a := \langle \Delta - d, x \rangle, \quad \nu := \sum_{e \in E} \vartheta_e f_e(x)^2.$$

By definition of D ,

$$D(x) = a - \frac{1}{2}\nu - \frac{\sigma}{2}x^\top D x.$$

Since $x^\top D x \geq 0$, we have

$$D(x) \leq a - \frac{1}{2}\nu.$$

Hence

$$a - \frac{1}{2}\nu \geq D(x) > 0.$$

In particular, $a > 0$.

By K -sweep locality,

$$\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e) \leq K\sqrt{\nu}.$$

Dividing by $a > 0$, we get

$$\frac{\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e)}{a} \leq K \frac{\sqrt{\nu}}{a}.$$

We now bound $\sqrt{\nu}/a$. Since $a - \nu/2 > 0$,

$$\frac{\sqrt{\nu}}{a} \leq \frac{1}{\sqrt{2(a - \nu/2)}}.$$

Indeed, after squaring both sides, this inequality is equivalent to

$$2\nu(a - \nu/2) \leq a^2,$$

which is equivalent to

$$(a - \nu)^2 \geq 0.$$

Using $a - \nu/2 \geq D(x)$, we obtain

$$\frac{\sqrt{\nu}}{a} \leq \frac{1}{\sqrt{2D(x)}}.$$

Therefore

$$\frac{\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e)}{a} \leq \frac{K}{\sqrt{2D(x)}}. \tag{5}$$

For $h > 0$, define the sweep set

$$S_h := \{v \in V : x_v \geq h\}.$$

Also define

$$q(h) := \Delta(S_h) - \text{vol}(S_h),$$

and

$$r(h) := \sum_{e \in E} \vartheta_e w_e(S_h \cap e) \ell_b(e).$$

For nonnegative x , the standard layer-cake identities give

$$a = \int_0^\infty q(h) dh,$$

and the Lovász coarea formula gives

$$\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e) = \int_0^\infty r(h) dh.$$

Let

$$H_+ := \{h > 0 : q(h) > 0\}.$$

Since

$$a = \int_0^\infty q(h) dh > 0,$$

the set H_+ is nonempty. Moreover, because $q(h) \leq 0$ outside H_+ ,

$$\int_{H_+} q(h) dh \geq a.$$

Also $r(h) \geq 0$, so

$$\int_{H_+} r(h) dh \leq \int_0^\infty r(h) dh.$$

Thus,

$$\frac{\int_{H_+} r(h) dh}{\int_{H_+} q(h) dh} \leq \frac{\int_0^\infty r(h) dh}{a}.$$

By averaging, there exists some $h \in H_+$ such that

$$\frac{r(h)}{q(h)} \leq \frac{\int_{H_+} r(h) dh}{\int_{H_+} q(h) dh} \leq \frac{\int_0^\infty r(h) dh}{a}.$$

Using (5),

$$\frac{r(h)}{q(h)} \leq \frac{K}{\sqrt{2D(x)}}. \quad (6)$$

Since $h \in H_+$, we have

$$q(h) > 0.$$

By definition of $\ell_b(e)$,

$$\ell_b(e) \geq \frac{1}{\sqrt{\text{vol}(C)}}.$$

Therefore

$$\begin{aligned} r(h) &= \sum_{e \in E} \vartheta_e w_e(S_h \cap e) \ell_b(e) \\ &\geq \frac{1}{\sqrt{\text{vol}(C)}} \sum_{e \in E} \vartheta_e w_e(S_h \cap e) \\ &= \frac{\text{vol}(\partial S_h)}{\sqrt{\text{vol}(C)}}. \end{aligned}$$

On the other hand,

$$q(h) = \Delta(S_h) - \text{vol}(S_h) \leq \Delta(S_h).$$

Since $\Delta(v) = \delta d_v$ for $v \in S$ and $\Delta(v) = 0$ otherwise,

$$\Delta(S_h) = \delta \text{vol}(S_h \cap S) \leq \delta \text{vol}(S_h).$$

Using $\delta = 3/\alpha$, we get

$$q(h) \leq \frac{3}{\alpha} \text{vol}(S_h).$$

Combining the previous inequalities with equation 6 gives,

$$\frac{\text{vol}(\partial S_h)}{\sqrt{\text{vol}(C)}} \leq r(h) \leq \frac{K}{\sqrt{2D(x)}} q(h) \leq \frac{K}{\sqrt{2D(x)}} \cdot \frac{3}{\alpha} \text{vol}(S_h).$$

Hence

$$\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3K}{\alpha} \sqrt{\frac{\text{vol}(C)}{2D(x)}}. \quad (7)$$

By Assumption 2,

$$\text{vol}(C) \leq \text{vol}(V \setminus C),$$

so

$$\Phi(C) = \frac{\text{vol}(\partial C)}{\text{vol}(C)}.$$

Therefore

$$D_C = \frac{\text{vol}(C)^2}{2 \text{vol}(\partial C)} = \frac{\text{vol}(C)}{2\Phi(C)}.$$

Equivalently,

$$\text{vol}(C) = 2\Phi(C)D_C.$$

Substituting this identity into equation 7 gives

$$\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3K}{\alpha} \sqrt{\Phi(C)} \sqrt{\frac{D_C}{D(x)}}.$$

Finally,

$$D(x) \geq D(x^\star) - \varepsilon \geq D_C - \varepsilon = D_C \left(1 - \frac{\varepsilon}{D_C}\right).$$

Thus

$$\sqrt{\frac{D_C}{D(x)}} \leq \frac{1}{\sqrt{1 - \varepsilon/D_C}}.$$

Consequently,

$$\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3K}{\alpha} \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1 - \varepsilon/D_C}}.$$

If additionally

$$\text{vol}(S_h) \leq \text{vol}(V \setminus S_h),$$

then

$$\Phi(S_h) = \frac{\text{vol}(\partial S_h)}{\min\{\text{vol}(S_h), \text{vol}(V \setminus S_h)\}} = \frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)}.$$

Hence the same bound holds for $\Phi(S_h)$. □

B.7 Proof of Corollary 2: Early stopping clustering guarantee for TL-HFD

Auxiliary Lemma 4 (Support-volume sufficient condition for sweep locality). *Let $x \geq 0$, and let $A_x := \text{supp}(x) = \{v \in \mathcal{V} : x_v > 0\}$. Then x satisfies $\mathbb{K}$ -sweep locality with $\mathbb{K} = 1 + \sqrt{\frac{\text{vol}(A_x)}{\text{vol}(C)}}$. In particular, if $\text{vol}(A_x) \leq \left(\frac{4}{\beta} - 1\right)^2 \text{vol}(C)$, then x satisfies the original HFD sweep-locality condition $\sum_{e \in \mathcal{E}} \vartheta_e f_e(x) \ell_b(e) \leq \frac{4}{\beta} \sqrt{\nu(x)}$.*

Proof. If $\nu(x) = 0$, then $f_e(x) = 0$ for every e with $\vartheta_e > 0$. Therefore

$$\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e) = 0 = \left(1 + \sqrt{\frac{\text{vol}(A_x)}{\text{vol}(C)}}\right) \sqrt{\nu(x)}.$$

Thus the claim is trivial in this case. We may henceforth assume $\nu(x) > 0$.

Let

$$E_x := \{e \in \mathcal{E} : f_e(x) > 0\}.$$

If $e \cap A_x = \emptyset$, then x is identically zero on e , so $x|_e \equiv 0$. Since the cut-costs are normalized, $0 \in B_e$, and hence $f_e(0) = 0$. Therefore $f_e(x) = 0$ whenever $e \cap A_x = \emptyset$. Thus every hyperedge in E_x intersects A_x .

Consequently,

$$\sum_{e \in E_x} \vartheta_e \leq \sum_{e: e \cap A_x \neq \emptyset} \vartheta_e.$$

Each hyperedge e with $e \cap A_x \neq \emptyset$ is counted at least once in

$$\sum_{v \in A_x} \sum_{e \ni v} \vartheta_e.$$

Therefore

$$\sum_{e \in E_x} \vartheta_e \leq \sum_{v \in A_x} \sum_{e \ni v} \vartheta_e = \sum_{v \in A_x} d_v = \text{vol}(A_x).$$

Next, since

$$\ell_b(e) = \max \left\{ \frac{1}{\sqrt{\text{vol}(C)}}, \frac{f_e(x)}{\sqrt{\nu(x)}} \right\},$$

we have the elementary upper bound

$$\ell_b(e) \leq \frac{1}{\sqrt{\text{vol}(C)}} + \frac{f_e(x)}{\sqrt{\nu(x)}}.$$

Thus

$$\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e) \leq \frac{1}{\sqrt{\text{vol}(C)}} \sum_{e \in E} \vartheta_e f_e(x) + \frac{1}{\sqrt{\nu(x)}} \sum_{e \in E} \vartheta_e f_e(x)^2.$$

The second term is exactly

$$\frac{1}{\sqrt{\nu(x)}} \sum_{e \in E} \vartheta_e f_e(x)^2 = \sqrt{\nu(x)}.$$

For the first term, since $f_e(x) = 0$ outside E_x ,

$$\sum_{e \in E} \vartheta_e f_e(x) = \sum_{e \in E_x} \vartheta_e f_e(x).$$

By Cauchy–Schwarz,

$$\begin{aligned} \sum_{e \in E_x} \vartheta_e f_e(x) &\leq \left(\sum_{e \in E_x} \vartheta_e \right)^{1/2} \left(\sum_{e \in E_x} \vartheta_e f_e(x)^2 \right)^{1/2} \\ &\leq \sqrt{\text{vol}(A_x)} \sqrt{\nu(x)}. \end{aligned}$$

Substituting this bound gives

$$\begin{aligned} \sum_{e \in E} \vartheta_e f_e(x) \ell_b(e) &\leq \frac{\sqrt{\text{vol}(A_x)}}{\sqrt{\text{vol}(C)}} \sqrt{\nu(x)} + \sqrt{\nu(x)} \\ &= \left(1 + \sqrt{\frac{\text{vol}(A_x)}{\text{vol}(C)}}\right) \sqrt{\nu(x)}. \end{aligned}$$

Hence x satisfies $\mathbb{K}$ -sweep locality with

$$\mathbb{K} = 1 + \sqrt{\frac{\text{vol}(A_x)}{\text{vol}(C)}}.$$

Finally, if

$$\text{vol}(A_x) \leq \left(\frac{4}{\beta} - 1\right)^2 \text{vol}(C),$$

then

$$1 + \sqrt{\frac{\text{vol}(A_x)}{\text{vol}(C)}} \leq \frac{4}{\beta}.$$

Therefore

$$\sum_{e \in E} \vartheta_e f_e(x) \ell_b(e) \leq \frac{4}{\beta} \sqrt{\nu(x)},$$

which is the original HFD sweep-locality condition. $\square$

Corollary 2 (Early-stopping sweep guarantee for TL-HFD). *Assume Assumptions 1, 2, and 3. Assume also the hypotheses of Corollary 1, using the box-constrained Proposition 1 so that $\varepsilon_{\text{vol}}(T)$ is well-defined. Let $\hat{x}^{(T)} \in \arg \min_{0 \leq t < T} F(x^{(t)})$, and define $\zeta_T := \text{vol}(\text{supp}(\hat{x}^{(T)})) / \text{vol}(C)$. If $\varepsilon_{\text{vol}}(T) < D_C := \frac{\text{vol}(C)^2}{2 \text{vol}(\partial C)}$, then there exists a threshold $h > 0$ such that $S_h := \{v : \hat{x}_v^{(T)} \geq h\}$ satisfies $\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3}{\alpha} (1 + \sqrt{\zeta_T}) \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1 - \varepsilon_{\text{vol}}(T)/D_C}}$. If additionally $\text{vol}(S_h) \leq \text{vol}(\mathcal{V} \setminus S_h)$, then the same bound holds for $\Phi(S_h)$.*

Proof. By Corollary 1, given in Appendix B.5, with the algorithm-scope clarification at the start of its proof, the $\varepsilon_{\text{vol}}(T)$ bound is a formal guarantee for the box-constrained Algorithm 2. The same bound also applies to any run of the unconstrained Algorithm 1 whose iterates remain below the box ceiling, since then the two algorithms coincide; see Remark 7.

By Corollary 1, under the box-constrained Proposition 1, the best iterate satisfies

$$F(\hat{x}^{(T)}) - F(x^*) \leq \varepsilon_{\text{vol}}(T).$$

Since $\varepsilon_{\text{vol}}(T) < D_C$ by assumption, the suboptimality requirement in Theorem 2 is satisfied with $x = \hat{x}^{(T)}$ and $\varepsilon = \varepsilon_{\text{vol}}(T)$.

It remains to verify the K -sweep-locality condition. Let $A_{\hat{x}} := \text{supp}(\hat{x}^{(T)})$. By definition,

$$\zeta_T = \frac{\text{vol}(A_{\hat{x}})}{\text{vol}(C)}.$$

Therefore, by Auxiliary Lemma 4, $\hat{x}^{(T)}$ satisfies K_T -sweep locality with

$$K_T = 1 + \sqrt{\frac{\text{vol}(A_{\hat{x}})}{\text{vol}(C)}} = 1 + \sqrt{\zeta_T}.$$

Applying Theorem 2 with $x = \hat{x}^{(T)}$, $\varepsilon = \varepsilon_{\text{vol}}(T)$, and $K = K_T$, there exists a threshold $h > 0$ such that $S_h := \{v : \hat{x}_v^{(T)} \geq h\}$ satisfies

$$\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3K_T}{\alpha} \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1 - \varepsilon_{\text{vol}}(T)/D_C}}.$$

Substituting $K_T = 1 + \sqrt{\zeta_T}$ gives

$$\frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)} \leq \frac{3}{\alpha} (1 + \sqrt{\zeta_T}) \sqrt{\Phi(C)} \cdot \frac{1}{\sqrt{1 - \varepsilon_{\text{vol}}(T)/D_C}}.$$

If additionally $\text{vol}(S_h) \leq \text{vol}(V \setminus S_h)$, then

$$\Phi(S_h) = \frac{\text{vol}(\partial S_h)}{\min\{\text{vol}(S_h), \text{vol}(V \setminus S_h)\}} = \frac{\text{vol}(\partial S_h)}{\text{vol}(S_h)}.$$

Hence the same bound holds for $\Phi(S_h)$. □

C Explored-Volume Bound (Theorem 3)

This section provides the explicit local ingredients needed for the B -based truncation bound and for the activated-volume certificate. In particular, Proposition 1 gives an explicit bound on the restricted local subgradient norm under the box-constrained variant, and Theorem 3 certifies the activated volume during the run.

C.1 Bounding the optimal solution

Auxiliary Lemma 5 (Optimal solution bound). *Let $x^* \in \arg \min_{x \geq 0} F(x)$ with $\delta > 1$ and $\sigma > 0$. Then*

$$\|x^*\|_\infty \leq \frac{\delta - 1}{\sigma}.$$

Proof. Let

$$m := \|x^*\|_\infty, \quad U := \{v \in V : x_v^* = m\}.$$

If $m = 0$, then the claim is immediate. Assume henceforth that $m > 0$.

Since x^* minimizes F over $\mathbb{R}_+^{|\mathcal{V}|}$, the KKT conditions imply that there exists a subgradient $g^* \in \partial F(x^*)$ such that

$$g_v^* = 0 \quad \forall v \in U,$$

because every $v \in U$ satisfies $x_v^* = m > 0$.

By the subgradient form of Lemma 1,

$$g^* = \sum_{e \in E} \vartheta_e f_e(x^*) \rho_e(x^*) + \sigma D x^* - (\Delta - d),$$

for some choice $\rho_e(x^*) \in \partial f_e(x^*)$ for each $e \in E$. Summing over U , we obtain

$$\begin{aligned} 0 &= \sum_{v \in U} g_v^* \\ &= \sum_{e \in E} \vartheta_e f_e(x^*) \sum_{v \in U \cap e} [\rho_e(x^*)]_v + \sigma \sum_{v \in U} d_v x_v^* - \Delta(U) + \text{vol}(U). \end{aligned}$$

Since $x_v^* = m$ for all $v \in U$,

$$\sum_{v \in U} d_v x_v^* = m \text{vol}(U).$$

Hence

$$0 = \sum_{e \in E} \vartheta_e f_e(x^*) \rho_e(x^*)(U \cap e) + \sigma m \text{vol}(U) - \Delta(U) + \text{vol}(U),$$

where

$$\rho_e(x^*)(U \cap e) := \sum_{v \in U \cap e} [\rho_e(x^*)]_v.$$

We now show that each edge contribution is nonnegative. Fix $e \in E$, and let $U_e := U \cap e$. Since every vertex in U_e attains the maximum value of x^* on e , U_e is precisely the top level-set block of $x^*|_e$. By the standard characterization of Lovász subgradients, every $\rho_e(x^*) \in \partial f_e(x^*)$ is a convex combination of greedy extreme subgradients associated with permutations consistent with the ordering of $x^*|_e$. For each such greedy extreme subgradient, the total mass on the first block U_e is

$$w_e(U_e) - w_e(\emptyset) = w_e(U_e).$$

Therefore, for every $\rho_e(x^*) \in \partial f_e(x^*)$,

$$\rho_e(x^*)(U_e) = w_e(U_e) \geq 0,$$

using the normalization $w_e(\emptyset) = 0$ and nonnegativity of cut costs. Since also $f_e(x^*) \geq 0$, every summand

$$\vartheta_e f_e(x^*) \rho_e(x^*)(U_e)$$

is nonnegative. Thus

$$\sum_{e \in E} \vartheta_e f_e(x^*) \rho_e(x^*)(U \cap e) \geq 0.$$

It follows that

$$0 \geq \sigma m \operatorname{vol}(U) - \Delta(U) + \operatorname{vol}(U),$$

or equivalently

$$\sigma m \operatorname{vol}(U) \leq \Delta(U) - \operatorname{vol}(U).$$

Now

$$\Delta(U) = \delta \operatorname{vol}(U \cap S),$$

so

$$\Delta(U) - \operatorname{vol}(U) = \delta \operatorname{vol}(U \cap S) - \operatorname{vol}(U) \leq (\delta - 1) \operatorname{vol}(U),$$

because $\operatorname{vol}(U \cap S) \leq \operatorname{vol}(U)$. Therefore

$$\sigma m \operatorname{vol}(U) \leq (\delta - 1) \operatorname{vol}(U).$$

Since $U \neq \emptyset$, we have $\operatorname{vol}(U) > 0$, and hence

$$m \leq \frac{\delta - 1}{\sigma}.$$

As $m = \|x^*\|_\infty$, the claim follows. $\square$

C.2 Box-constrained variant for explicit constants

The localized convergence theorem does not require bounded iterates. However, to obtain the explicit coordinatewise bound used in Proposition 1 and the B -based forms of Corollaries 1 and 2, we introduce a box-constrained theoretical variant. By Auxiliary Lemma 5, the optimizer already satisfies $\|x^*\|_\infty \leq (\delta - 1)/\sigma$, so this clipping does not change the optimum; it is used only to keep intermediate iterates uniformly bounded.

Define $\bar{x} := (\delta - 1)/\sigma$ and the box

$$\mathcal{X} := \{x \in \mathbb{R}^V : 0 \leq x_v \leq \bar{x} \ \forall v \in V\}.$$

By Auxiliary Lemma 5, $x^* \in \mathcal{X}$, so $\arg \min_{x \in \mathcal{X}} F(x) = x^*$. We modify the algorithm 1 by replacing the projection $\Pi_{\mathbb{R}_+^V}$ with the coordinatewise box projection $[\Pi_{\mathcal{X}}(z)]_v = \min(\max(z_v, 0), \bar{x})$ as given in Algorithm 2.

Auxiliary Lemma 6 (Box projection preserves locality). *The box-constrained variant preserves all properties stated in Lemma 2. In particular, for every $u \notin A^{(t)} \cup \partial A^{(t)}$,*

$$[\Pi_{\mathcal{X}}(x^{(t)} - \eta D^{-1}g^{(t)})]_u = 0.$$

Moreover, $\|x^{(t)}\|_\infty \leq \bar{x}$ for all $t \geq 0$.

Proof. For $u \notin A^{(t)} \cup \partial A^{(t)}$: $x_u^{(t)} = 0$ and $g_u = d_u > 0$ (Lemma 2), so $y_u = -\eta < 0$ and $\Pi_{\mathcal{X}}(y_u) = 0$. The iterate bound $\|x^{(t)}\|_\infty \leq \bar{x}$ follows by construction. Nonexpansiveness of $\Pi_{\mathcal{X}}$ in $\|\cdot\|_D$ holds because $\mathcal{X}$ is a convex set and the projection is coordinatewise with D diagonal. $\square$

Remark 6. For the explicit local subgradient bound, we consider the box-constrained variant obtained by replacing projection onto $\mathbb{R}_+^{|V|}$ with projection onto $\mathcal{X} = \{x : 0 \leq x_v \leq (\delta - 1)/\sigma\}$. This does not change the optimizer, since $\|x^*\|_\infty \leq (\delta - 1)/\sigma$.

Algorithm 2 Thresholded Local Hyper-Flow Diffusion (TL-HFD, box-constrained theoretical variant)

Require: Current iterate $x^{(t)}$, seed set S , step size η_{t+1} , activation count $k \geq 0$, box radius $\bar{x} = (\delta - 1)/\sigma$

Ensure: Next iterate $x^{(t+1)}$

Active region and its one-hop boundary.

- 1: $A^{(t)} \leftarrow \text{supp}(x^{(t)}) \cup S$
- 2: $\partial A^{(t)} \leftarrow \{u \notin A^{(t)} : \exists e \in E, u \in e, e \cap A^{(t)} \neq \emptyset\}$

Compute subgradient only on the local region (Lemma 2).

- 3: $g^{(t)} \leftarrow g(x^{(t)})$ restricted to $A^{(t)} \cup \partial A^{(t)}$
- 4: $x^{(t+1)} \leftarrow x^{(t)}$

Box-projected PSGD update on active vertices.

- 5: **for all** $u \in A^{(t)}$ **do**
- 6: $x_u^{(t+1)} \leftarrow \min \left\{ \bar{x}, \max \left\{ 0, x_u^{(t)} - \eta_{t+1}[g^{(t)}]_u/d_u \right\} \right\}$
- 7: **end for**

Scoring for boundary vertices.

- 8: **for all** $u \in \partial A^{(t)}$ **do**
- 9: $\kappa^{(t)}(u) \leftarrow \max \{0, -[g^{(t)}]_u/d_u\}$
- 10: $c^{(t)}(u) \leftarrow \left(d_{\text{in}}^{(t)}(u)/d(u) \right)^\gamma$
- 11: $s^{(t)}(u) \leftarrow \kappa^{(t)}(u) c^{(t)}(u)$
- 12: **end for**

Activate the top-k boundary vertices.

- 13: $K_t \leftarrow \text{TOPK}(s^{(t)}, k)$
- 14: **for all** $u \in K_t$ **do**
- 15: $x_u^{(t+1)} \leftarrow \min \{ \bar{x}, \eta_{t+1}\kappa^{(t)}(u) \}$
- 16: **end for**
- 17: **return** $x^{(t+1)}$

Remark 7 (Box constraint is non-binding in practice). Although Proposition 1 is stated for the box-constrained variant, Algorithm 2, our experiments use the unconstrained Algorithm 1. The two algorithms produce identical iterates whenever no coordinate of the unconstrained iterate reaches the ceiling $\bar{x} = (\delta - 1)/\sigma$. This is the case in our experiments. For example, for Amazon runs with $\delta_{\text{exp}} = 200$ and $\sigma = 10^{-4}$, the seed-specific theoretical coefficient is $\delta(s^*) = \delta_{\text{exp}} \text{vol}(T)/d_{s^*}$, so $\bar{x} = ((\delta(s^*) - 1)/\sigma) \geq 1.99 \times 10^6$. The observed coordinates of $x^{(t)}$ remain below 10^2 throughout all runs. Hence the box projection is never active on these instances, and the iterates of Algorithm 1 and Algorithm 2 coincide. The box constraint therefore plays a theoretical role: it yields the uniform coordinatewise bound $|g_v^{(t)}/d_v| \leq B_{\text{loc}}$ established in Proposition 1 and used in the convergence corollaries, without affecting the empirical behavior of the algorithm. Moreover, the box projection $\Pi_{\mathcal{X}}$ is coordinatewise and 1-Lipschitz in the D -norm (Auxiliary Lemma 6). For skipped boundary coordinates, the box-projected truncation error is at most $\eta_{t+1}\kappa^{(t)}(u)$, so the same truncation-error upper bound used in Lemma 3 applies to Algorithm 2.

C.3 Explicit bound on the local subgradient norm

Proposition 1 (Explicit local subgradient bound). *Under the box-constrained algorithm with $\|x^{(t)}\|_\infty \leq \bar{x} = (\delta - 1)/\sigma$, $\delta \geq 2$, and the cut-cost normalization $0 \leq w_e(S) \leq 1$, for every iteration t :*

$$\|g^{(t)}\|_{D^{-1}, L_t}^2 \leq B^2 \cdot \text{vol}(L_t), \quad B := (\delta - 1) \left(\frac{1}{\sigma} + 2 \right).$$

Proof. It suffices to show $|g_v/d_v| \leq B$ for all $v \in L_t$, since this implies $g_v^2/d_v \leq B^2 d_v$.

Bounding the Lovász contribution. We first establish that for every $v \in L_t$,

$$\frac{1}{d_v} \left| \sum_{e \ni v} \vartheta_e f_e(x^{(t)}) [\rho_e(x^{(t)})]_v \right| \leq \bar{x}. \quad (8)$$

Since $\rho_e \in B_e$ and $0 \leq w_e(S) \leq 1$, for any single vertex $v \in e$ we have

$$[\rho_e]_v = \langle \rho_e, \mathbf{1}_{\{v\}} \rangle \leq w_e(\{v\}) \leq 1,$$

and

$$[\rho_e]_v = \langle \rho_e, \mathbf{1}_e \rangle - \langle \rho_e, \mathbf{1}_{e \setminus \{v\}} \rangle = 0 - \langle \rho_e, \mathbf{1}_{e \setminus \{v\}} \rangle \geq -w_e(e \setminus \{v\}) \geq -1,$$

so $|[\rho_e]_v| \leq 1$. Since $f_e(x) \geq 0$, this gives $|\sum_{e \ni v} \vartheta_e f_e(x) [\rho_e]_v| \leq \sum_{e \ni v} \vartheta_e f_e(x)$.

For $x \geq 0$ with $\|x\|_\infty \leq \bar{x}$: since $\langle \rho, \mathbf{1}_S \rangle \leq 1$ for all $S \subseteq e$, the positive components of any $\rho \in B_e$ sum to at most 1. Therefore $f_e(x) = \max_{\rho \in B_e} \langle \rho, x \rangle \leq \|x\|_\infty \leq \bar{x}$, and

$$\sum_{e \ni v} \vartheta_e f_e(x) \leq \bar{x} \sum_{e \ni v} \vartheta_e = \bar{x} d_v,$$

establishing (8).

Bounding $|g_v/d_v|$ by cases. Recall that $g_v/d_v = [\text{Lov}]_v/d_v + \sigma x_v - (\Delta(v) - d_v)/d_v$, where $[\text{Lov}]_v := \sum_{e \ni v} \vartheta_e f_e(x) [\rho_e]_v$.

Case (a): $v \in S \cap A^{(t)}$ (seed vertex). Here $\Delta(v) = \delta d_v$, so $(\Delta(v) - d_v)/d_v = \delta - 1$. Using (8) and $x_v \leq \bar{x}$:

$$\left| \frac{g_v}{d_v} \right| \leq \bar{x} + \sigma \bar{x} + (\delta - 1) = (1 + \sigma) \frac{\delta - 1}{\sigma} + (\delta - 1) = (\delta - 1) \left(\frac{1}{\sigma} + 2 \right) = B.$$

Case (b): $v \in A^{(t)} \setminus S$ (active, non-seed). Here $\Delta(v) = 0$, so $(\Delta(v) - d_v)/d_v = -1$.

$$\left| \frac{g_v}{d_v} \right| \leq \bar{x} + \sigma \bar{x} + 1 = \frac{\delta - 1}{\sigma} + (\delta - 1) + 1 = \frac{\delta - 1}{\sigma} + \delta.$$

Case (c): $v \in \partial A^{(t)}$ (boundary). Here $x_v = 0$ and $\Delta(v) = 0$.

$$\left| \frac{g_v}{d_v} \right| \leq \bar{x} + 0 + 1 = \frac{\delta - 1}{\sigma} + 1.$$

For $\delta \geq 2$ (guaranteed since $\delta = 3/\alpha \geq 3$), Case (a) dominates:

- Case (a) $\geq$ Case (b): $(\delta - 1)(1/\sigma + 2) \geq (\delta - 1)/\sigma + \delta$ iff $2(\delta - 1) \geq \delta$ iff $\delta \geq 2$.
- Case (a) $\geq$ Case (c): $(\delta - 1)(1/\sigma + 2) \geq (\delta - 1)/\sigma + 1$ iff $2(\delta - 1) \geq 1$.

Thus $|g_v/d_v| \leq B$ for all $v \in L_t$, and therefore

$$\|g^{(t)}\|_{D^{-1}, L_t}^2 = \sum_{v \in L_t} \frac{g_v^2}{d_v} \leq B^2 \sum_{v \in L_t} d_v = B^2 \cdot \text{vol}(L_t). \quad \square$$

C.4 Localized explored-volume bound

Theorem 3 (Explored-volume). Assume $x^{(0)} = 0$. Let $M_0 := S$ and let $J_t := \{u \in \partial A^{(t)} : u \notin M_t, x_u^{(t+1)} > 0\}$ be the set of vertices activated for the first time at iteration t , where $M_{t+1} := M_t \cup J_t$. Equivalently, for boundary updates in Algorithm 1, $J_t = \{u \in K_t : u \notin M_t, \kappa^{(t)}(u) > 0\}$. Let $L_t := A^{(t)} \cup \partial A^{(t)}$ and $\kappa^{(t)}(u) := \max \left\{ 0, -\frac{g_u^{(t)}}{d_u} \right\}$. Let $I_T := \{0 \leq t < T : J_t \neq \emptyset\}$. Assume that for each $t \in I_T$, there exists $\rho_t > 0$ such that $\kappa^{(t)}(u) \geq \rho_t \forall u \in J_t$. Then, for $M_T = S \cup \bigcup_{t=0}^{T-1} J_t$, we have $\text{vol}(M_T) \leq \text{vol}(S) + \sum_{t \in I_T} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\rho_t^2}$. In particular, if $\|g^{(t)}\|_{D^{-1}, L_t} \leq G_{\text{loc}} \forall t$, then $\text{vol}(M_T) \leq \text{vol}(S) + G_{\text{loc}}^2 \sum_{t \in I_T} \rho_t^{-2}$.

Proof. Fix an iteration t . By construction of Algorithm 1, $x^{(0)} = 0$ and let $J_t := \{u \in \partial A^{(t)} : u \notin M_t, x_u^{(t+1)} > 0\}$ be the set of vertices activated for the first time at iteration t , where $M_{t+1} := M_t \cup J_t$. As every first-time activated vertex is selected from the current boundary, we get

$$J_t \subseteq \partial A^{(t)} \subseteq L_t.$$

By hypothesis,

$$\kappa^{(t)}(u) \geq \rho_t \quad \forall u \in J_t.$$

Therefore

$$\rho_t^2 \text{vol}(J_t) = \rho_t^2 \sum_{u \in J_t} d_u \leq \sum_{u \in J_t} d_u (\kappa^{(t)}(u))^2.$$

Since

$$\kappa^{(t)}(u) = \max \left\{ 0, -\frac{g_u^{(t)}}{d_u} \right\},$$

we have, for every u ,

$$d_u (\kappa^{(t)}(u))^2 \leq \frac{(g_u^{(t)})^2}{d_u}.$$

Thus, using $J_t \subseteq L_t$,

$$\rho_t^2 \text{vol}(J_t) \leq \sum_{u \in J_t} \frac{(g_u^{(t)})^2}{d_u} \leq \sum_{v \in L_t} \frac{(g_v^{(t)})^2}{d_v} = \|g^{(t)}\|_{D^{-1}, L_t}^2.$$

Hence

$$\text{vol}(J_t) \leq \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\rho_t^2}.$$

The sets J_t are pairwise disjoint because each vertex can be activated for the first time in at most one iteration. Moreover, $S \cap J_t = \emptyset$ for all t , since $S \subseteq A^{(0)}$. Since $\text{vol}(J_t) = 0$ for $t \notin I_t$ we restrict the sum to $t \in I_T$, where $\rho_t > 0$. Therefore

$$\text{vol}(M_T) = \text{vol}(S) + \sum_{t \in I_T} \text{vol}(J_t).$$

Substituting the preceding bound on $\text{vol}(J_t)$ gives

$$\text{vol}(M_T) \leq \text{vol}(S) + \sum_{t \in I_T} \frac{\|g^{(t)}\|_{D^{-1}, L_t}^2}{\rho_t^2}.$$

If additionally

$$\|g^{(t)}\|_{D^{-1}, L_t} \leq G_{\text{loc}} \quad \forall t,$$

then

$$\text{vol}(M_T) \leq \text{vol}(S) + G_{\text{loc}}^2 \sum_{t \in I_T} \rho_t^{-2}.$$

□

C.5 Known Facts: A technical fact about projections in the $\|\cdot\|_D$ norm

Equation 4 uses the coordinatewise Euclidean projection onto the nonnegative orthant, $\Pi_{\mathbb{R}_+^V}(z) = \max\{z, 0\}$ applied coordinatewise. Because the orthant constraint is separable across coordinates, this coincides with the projection in any diagonal weighted Euclidean norm (in particular, the $\|\cdot\|_D$ norm).

Lemma 4 (Projection in $\|\cdot\|_D$ is coordinatewise clipping). *Let $D = \text{diag}(d)$ with $d_u > 0$ for all u . Define the D -projection onto $\mathbb{R}_+^V$ by*

$$\Pi_{\mathbb{R}_+^V}^D(z) \in \arg \min \{\|x - z\|_D^2 : x \geq 0\}.$$

Then $\Pi_{\mathbb{R}_+^V}^D(z) = \Pi_{\mathbb{R}_+^V}(z)$ coordinatewise, i.e. $(\Pi_{\mathbb{R}_+^V}^D(z))_u = \max\{z_u, 0\}$. Moreover, $\Pi_{\mathbb{R}_+^V}$ is nonexpansive in $\|\cdot\|_D$:

$$\|\Pi_{\mathbb{R}_+^V}^D(a) - \Pi_{\mathbb{R}_+^V}^D(b)\|_D \leq \|a - b\|_D \quad \forall a, b \in \mathbb{R}^V.$$

Proof. Because D is diagonal, the projection problem decouples coordinatewise:

$$\min_{x \geq 0} \sum_{u \in V} d_u (x_u - z_u)^2 = \sum_{u \in V} \min_{x_u \geq 0} d_u (x_u - z_u)^2.$$

Each scalar problem has the unique minimizer $x_u^* = \max\{z_u, 0\}$, hence $\Pi_{\mathbb{R}_+^V}^D(z) = \Pi_{\mathbb{R}_+^V}(z)$.

For nonexpansiveness, fix a, b and set $p = \Pi_{\mathbb{R}_+^V}(a)$, $q = \Pi_{\mathbb{R}_+^V}(b)$. The scalar map $t \mapsto \max\{t, 0\}$ is 1-Lipschitz, so $|p_u - q_u| \leq |a_u - b_u|$ for each u . Multiplying by d_u and summing yields

$$\|p - q\|_D^2 = \sum_u d_u (p_u - q_u)^2 \leq \sum_u d_u (a_u - b_u)^2 = \|a - b\|_D^2.$$

Taking square roots gives the claim. $\square$

C.6 Directional derivatives and Lovász support functions (Known result - provided here for completeness)

Recall $f_e(x) = \max_{\rho \in B_e} \langle \rho, x \rangle$ and $M_e(x) = \arg \max_{\rho \in B_e} \langle \rho, x \rangle$.

Proposition 2 (Directional derivative formulas). *Fix $x \in \mathbb{R}^V$ and a hyperedge e .*

1. *For any direction h ,*

$$f'_e(x; h) = \max_{\rho \in M_e(x)} \langle \rho, h \rangle.$$

2. *Let $g_e(x) = \frac{1}{2} f_e(x)^2$. Then $g'_e(x; h) = f_e(x) f'_e(x; h)$.*

3. *Consequently, the directional derivative of F in equation (1) satisfies*

$$F'(x; h) = \sum_{e \in E} \vartheta_e f_e(x) \max_{\rho \in M_e(x)} \langle \rho, h \rangle + \sigma x^\top Dh - \langle \Delta - d, h \rangle.$$

Proof. (1) Define $\phi(t) := f_e(x + th) = \max_{\rho \in B_e} \langle \rho, x + th \rangle$. Because B_e is compact and the objective is continuous linear in ρ , the max is attained. Moreover ϕ is convex as a pointwise maximum of linear functions in t .

Lower bound: for any $\bar{\rho} \in M_e(x)$ and any $t > 0$,

$$f_e(x + th) \geq \langle \bar{\rho}, x + th \rangle = f_e(x) + t \langle \bar{\rho}, h \rangle,$$

so $\frac{f_e(x+th) - f_e(x)}{t} \geq \langle \bar{\rho}, h \rangle$. Taking $\max_{\bar{\rho} \in M_e(x)}$ and then $t \downarrow 0$ gives $f'_e(x; h) \geq \max_{\rho \in M_e(x)} \langle \rho, h \rangle$.

Upper bound: for $t > 0$ let $\rho_t \in M_e(x + th)$. Then

$$f_e(x + th) = \langle \rho_t, x + th \rangle = \langle \rho_t, x \rangle + t \langle \rho_t, h \rangle \leq f_e(x) + t \langle \rho_t, h \rangle,$$

since $\langle \rho_t, x \rangle \leq \max_{\rho \in B_e} \langle \rho, x \rangle = f_e(x)$. Thus $\frac{f_e(x+th) - f_e(x)}{t} \leq \langle \rho_t, h \rangle$. As $t \downarrow 0$, compactness implies $\{\rho_t\}$ has limit points, and any limit point must lie in $M_e(x)$ by continuity of $\rho \mapsto \langle \rho, x \rangle$. Taking $\limsup_{t \downarrow 0}$ yields $f'_e(x; h) \leq \max_{\rho \in M_e(x)} \langle \rho, h \rangle$. Combining proves (1).

(2) Let $\psi(t) := f_e(x + th)$. Since f_e is convex and finite everywhere, ψ is convex and hence directionally differentiable at 0. The scalar map $s \mapsto \frac{1}{2} s^2$ is differentiable with derivative s . Therefore, by the one-sided chain rule for directional derivatives,

$$g'_e(x; h) = \left(\frac{1}{2} \psi(t)^2 \right)'_{t=0+} = \psi(0) \psi'(0+) = f_e(x) f'_e(x; h).$$

(3) F is a finite sum of directionally differentiable functions, so the directional derivative is the sum of directional derivatives. Apply (2) to each e , and use the standard derivative of the quadratic and linear terms. $\square$

D Additional Experimental Details

D.1 Real World Dataset Statistics

Table 2 provides the statistics of the real-world datasets used for our empirical study. We now describe the datasets used together with the evaluation criteria.

Table 2: Summary statistics of datasets used in the experiments.

Dataset	$ V $	$ E $	Mean/Median	Clusters (#nodes)	Cut-costs
Trivago-clicks 4, D.5	172,738	233,202	5.6 / 2.0	10 (144–945)	Unit, Card
Amazon-reviews D.6	2,268,231	4,285,363	32.2 / 11.0	9 (31–5334)	Unit
Florida Bay D.7	126	141,233	4,483.6/ 3,770.5	3 (17–70)	Sub
High-school-contact D.8	327	7,818	55.6/ 53	9 (29–44)	Unit, Card
Oil trade (2017) D.9	229	100,639	1,757.9/175.0	— (ranking)	Sub

Trivago-Clicks [Chodrow et al., 2021]. The nodes in this hypergraph represent accommodations and hotels. A hyperedge connects a set of nodes when a user executes a “click-out” action during the same browsing session, indicating that the user was redirected to a partner website. The graph is constructed by employing geographical locations as ground truth cluster identities. The dataset contains 160 such clusters in total. Similar to [Fountoulakis et al., 2021], we focus on all clusters in this dataset that contain fewer than 1, 000 nodes and exhibit conductance below 0.25.

Amazon-reviews [Ni et al., 2019, Veldt et al., 2020]. This hypergraph is constructed from Amazon product review data, with each node representing a product. Products are connected by a hyperedge when they are reviewed by the same individual. Similar to Fountoulakis et al. [2021], the product category labels are employed as ground truth cluster identities. The dataset encompasses 29 product categories in total. Given our primary focus on local clustering, we examine all clusters containing fewer than 10, 000 nodes.

High-school-contact [Chodrow et al., 2021, Mastrandrea et al., 2015] Nodes in this hypergraph correspond to high school students. Students form a hyperedge when they are all detected in close proximity simultaneously, as measured by wearable sensors. The ground truth clusters are defined based on classroom assignments, with 9 classrooms represented in the dataset.

Florida Bay food network [Li and Milenkovic, 2017]. Each node in this hypergraph represents a species or organism inhabiting the Bay, while hyperedges encode transformed network motifs. Nodes are labeled according to their role in the food chain. Submodular cut costs are shown to be more appropriate for this dataset [Fountoulakis et al., 2021].

Oil Trade 2017 [Fountoulakis et al., 2021]. This hypergraph is constructed from 2017 international oil-trade records in the UN Comtrade dataset. Each node represents a country, and a 4-tuple $\{v_1, v_2, v_3, v_4\}$ forms a hyperedge if the trade surplus from each of v_1 and v_2 to each of v_3 and v_4 exceeds 10 million USD. This threshold is approximately the 80th percentile of country-level oil export values. Two countries are grouped in the same hyperedge when they share at least *two* major trading partners. Similar to Fountoulakis et al. [2021], we use this network for a node-ranking task rather than a clustering task. As with the Florida Bay food network, submodular cut-costs are more appropriate for this dataset.

D.2 Hypergraph cut-cost models

Our experiments use three cut-cost classes. Under *unit* cut-cost, every non-trivial split of a hyperedge incurs cost 1. Under *cardinality* cut-cost, the penalty depends on the number of nodes on each side: $w_e(S) = \min(|S \cap e|, |e \setminus S|)/\lfloor |e|/2 \rfloor$, giving finer-grained credit for edges that overlap heavily with the active set. Under *general submodular* cut-cost, an arbitrary submodular function w_e assigns a different penalty to each split, capturing directional or role-based structure

(e.g., prey–predator interactions in food webs). We prefix methods by U- (unit), C- (cardinality), and S- (submodular) to indicate the cut-cost used.

D.3 Methods and Parameter Settings

D.3.1 HFD baseline [Fountoulakis et al., 2021].

We set the same parameters for HFD as described in Fountoulakis et al. [2021]. We use $\sigma = 10^{-4}$ in all clustering experiments. The total seed-injection mass is set to a constant multiple of the target-cluster volume: $\|\Delta\|_1 = t \text{ vol}(\mathcal{T})$, where $\mathcal{T}$ denotes the target cluster and t is a dataset-dependent injection factor. For Amazon-reviews, $t = 200$ for the smaller categories $\{1, 2, 3, 12, 18\}$, and $t = 50$ for the larger categories $\{15, 17, 24, 25\}$. For Trivago-clicks and High-school-contact, $t = 3$. For the Florida Bay food network, t is set to cluster-specific values $t = 20, 10, 5$ for clusters $\{1, 2, 3\}$, respectively. The injection factor t controls how far the diffusion spreads from the seed set. Following prior HFD evaluations, t is chosen large enough so that the diffusion reaches a nontrivial portion of the target cluster while still penalizing excessive spread through the objective. All the mass is placed on the single seed: $\Delta(s) = \|\Delta\|_1$. For the Oil trade (2017) ranking experiment, we follow the HFD protocol of Fountoulakis et al. [2021]. The seed node is Iran, and the seed-injection mass is set to $\Delta[\text{seed}] = 500,000$. Unlike the clustering experiments, which use $\sigma = 10^{-4}$, the Oil trade ranking experiment uses $\sigma = 0.1$. We run HFD with the iteration budget of $T = 20$ for submodular cut-costs. Since this dataset is used for node ranking rather than clustering, we do not report F1 or conductance. Instead, countries are ranked by the HFD diffusion score σx_v , where x is the returned HFD diffusion vector.

LH [Liu et al., 2021] and ACL [Andersen et al., 2006] baselines. For the LH baselines, we use the parameter choices recommended by the authors of Liu et al. [2021]. For both *-LH-2.0 and *-LH-1.4, we set $\gamma = 0.1$, $\rho = 0.5$, $\kappa = cr$, where r is the ratio between the number of seed nodes and the target-cluster size, and c is a tuning constant. For Amazon-reviews, we use $c = 0.025$, as recommended in Liu et al. [2021]. For Trivago-clicks, and Florida Bay, we also use $c = 0.025$. For High-school-contact, $c = 0.25$ is chosen by tuning so that both *-LH-2.0 and *-LH-1.4 obtain competitive performance. For ACL, we use the same parameter choices as specified in Fountoulakis et al. [2021]. For the LH baselines, we set $\delta = 1$ for U-LH-*, and $\delta = \max_{e \in E} |e|$ for C-LH-*.

TL-HFD. TL-HFD (Algorithm 1) inherits the seed-injection protocol from the HFD baseline. The regularization parameter is fixed at $\sigma = 10^{-4}$ across all clustering experiments. Each trial uses a single-seed diffusion: for a chosen seed vertex $s \in T$, the injection vector Δ has exactly one nonzero entry,

$$\Delta(s) = \delta_{\text{exp}} \cdot \text{vol}(T), \quad \Delta(v) = 0 \quad \text{for } v \neq s,$$

where T is the target cluster and δ_{exp} is a dataset-dependent injection factor.

For Amazon-reviews and Trivago-clicks, we run multiple independent trials per cluster, each using a different vertex as the single seed. For contact-high-school, Florida Bay, and the synthetic HSBM, we use every target-cluster vertex once as the seed (yielding $|\mathcal{T}|$ trials per cluster). For oil trade, the seed is fixed to Iran. We report the median F1 across trials in the clustering experiments.

Injection factors. The injection factors δ_{exp} follow the standard HFD protocol of Fountoulakis et al. [2021]: $\delta_{\text{exp}} = 3$ for Trivago-clicks, contact-high-school, and synthetic HSBM; $\delta_{\text{exp}} = 200$ for the smaller Amazon categories $\{1, 2, 3, 12, 18\}$ and $\delta_{\text{exp}} = 50$ for the larger categories $\{15, 17, 24, 25\}$; cluster-specific values $\{20, 10, 5\}$ for Florida Bay (Pelagic, Detritus, Benthic respectively); and $\Delta(\text{Iran}) = 500,000$ for the oil trade ranking experiment.

Boundary activation. TL-HFD (Algorithm 1) additionally requires a score exponent γ and a boundary-activation rule. Boundary vertices are ranked by the score

$$s^{(t)}(u) = \kappa^{(t)}(u) \left(\frac{d_{\text{in}}^{(t)}(u)}{d(u)} \right)^\gamma,$$

where γ controls the strength of the structural commitment factor.

Two activation regimes: Activation-scale selection (auto- k) and grid-over- k . TL-HFD selects k , the number of boundary vertices activated per iteration, in one of two ways depending on the dataset.

Under *Activation-scale selection* (auto- k), the user specifies a grid of fractions f , and at runtime the integer activation count k is computed as a function of f and the seed-injection magnitude. This rule lets a single fraction grid generalize across clusters of different sizes without retuning k directly. Under *grid-over- k* , the user instead supplies k from a small set of integer values. In both regimes, the configuration achieving the lowest output conductance is reported; ties are broken by selecting the smallest f (auto- k) or smallest k (grid-over- k). For the oil-trade ranking task, where there is no labeled target cluster and conductance is undefined, the configuration is selected by qualitative inspection of the returned ranking.

The auto- k rule activates the top-scoring boundary vertices according to

$$k = \max\{1, \lfloor f \cdot \text{vol}_{\text{est}} \rfloor\} \quad (\text{unit cut-cost}), \quad k = \max\{1, \lfloor f \cdot n_{\text{est}} \rfloor\} \quad (\text{cardinality cut-cost}),$$

where

$$\text{vol}_{\text{est}} = \frac{\max_v \Delta(v)}{\delta_{\text{exp}}}, \quad n_{\text{est}} = \frac{\text{vol}_{\text{est}}}{\bar{d}},$$

and $\bar{d}$ is the mean vertex degree. Because each trial uses a single seed with $\max_v \Delta(v) = \delta_{\text{exp}} \cdot \text{vol}(\mathcal{T})$, the estimator simplifies to $\text{vol}_{\text{est}} = \text{vol}(\mathcal{T})$, so f has a clear interpretation as a per-iteration boundary expansion rate relative to the target volume. Smaller f yields a more conservative, locality-respecting diffusion; larger f expands faster but increases the risk of activating vertices outside the target cluster. The fraction is selected from a small per-cluster grid (Section D.3.1). This rule is well-behaved on graphs with moderate density and a clearly defined target cluster: the range $f \in [10^{-3}, 10^{-1}]$ then produces k values in $[1, |\mathcal{T}|]$, allowing a single per-dataset sweep to identify a near-optimal f . We use auto- k on Trivago-clicks, Amazon-reviews, contact-high-school, and synthetic HSBM.

Two regimes break the volume-based scaling and require grid-over- k instead. The first is *very dense graphs*, where the target volume is large relative to $|V|$: on Florida Bay, $\bar{d} \approx 4,484$ and a target cluster of 35 vertices has $\text{vol}(T) \approx 173,000$, so even $f = 10^{-3}$ produces $k > |V|$ and auto- k saturates instantly. The second is *node-ranking tasks without a target cluster*, where $\text{vol}(T)$ is undefined: the oil trade diffusion is run from a single fixed seed (Iran) for the purpose of ranking trading partners, with no labeled target community to scale against. In both regimes we sweep k over a small grid and report the configuration that produces the lowest output conductance (Florida Bay) or the most semantically coherent ranking (oil trade).

We use $\gamma = 0.5$ on Amazon-reviews and $\gamma = 1.0$ on Trivago-clicks and contact-high-school. The smaller exponent on Amazon makes the structural commitment factor less aggressive, so the ranking remains more responsive to the gradient push $\kappa^{(t)}(u)$. This was empirically more stable on Amazon, where product co-review hyperedges often contain bridge products spanning multiple categories. On Trivago-clicks and contact-high-school, the cluster boundaries are cleaner — co-clicked accommodations and within-classroom contact events are associated with a single cluster — so a sharper $\gamma = 1.0$ produces better recovery.

Table 3: Dataset-specific TL-HFD parameters for real-world experiments. T_{iter} denotes the number of TL-HFD iterations; two values are given when unit and cardinality cut-costs use different counts.

Dataset	Cut-costs	T_{iter} (unit / card)	γ	Activation
Trivago-clicks	Unit, Card	500 / 1000	1.0	auto- k via f
Amazon-reviews	Unit	1000 / —	0.5	auto- k via f
Contact-high-school	Unit, Card	1000 / 1000	1.0	auto- k via f
Florida Bay	Sub	500 / —	1.0	grid over k
Oil trade (2017)	Sub	20 / —	1.0	grid over k

Fraction and k -grid summary for real-world datasets. Here we summarize the grids used to select the boundary activation scale in the experiments. For sparse-to-moderate clustering datasets, TL-HFD uses a fraction f and converts it to an activation count k through the auto- k rule. For dense submodular clustering and ranking datasets, such as Florida Bay and Oil Trade, the volume-based scaling is less meaningful as explained above: in Florida Bay the extreme density can make the resulting k too large, while in Oil Trade there is no labeled target cluster volume because the task is ranking rather than clustering. For these datasets, we sweep k directly over a small integer grid.

The auto- k rule uses different normalizations for the two cut-cost classes: $k = \max\{1, \lfloor f \cdot \text{vol}_{\text{est}} \rfloor\}$ for unit cut-cost and $k = \max\{1, \lfloor f \cdot n_{\text{est}} \rfloor\}$ for cardinality cut-cost, where $n_{\text{est}} = \text{vol}_{\text{est}}/\bar{d}$ and $\bar{d}$ is the mean vertex degree. For the same f , the cardinality rule therefore activates approximately $\bar{d}$ -fold fewer boundary vertices per iteration than the unit rule. To explore comparable activation counts across cut-cost classes, the cardinality grid is extended to roughly $\bar{d}$ -fold larger fractions: on Trivago-clicks ($\bar{d} \approx 5.6$), this corresponds to unit fractions in $[0.01, 0.05]$ and cardinality fractions in $[0.01, 0.10]$. The same scaling motivates the corresponding grid for submodular cut-cost on datasets where it is used. This is purely a normalization difference in how f maps to k .

Table 4: Activation-scale grids used for TL-HFD. A fraction grid means that k is computed by the auto- k rule; a direct k -grid means that the integer activation count is swept directly.

Dataset	Cut-cost	Activation-scale grid
Trivago-clicks	unit	$f \in \{0.01, 0.02, 0.03, 0.05\}$
Trivago-clicks	cardinality	$f \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.10\}$
Amazon-reviews	unit	$f \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.03, 0.05, 0.07, 0.1\}$
Contact-high-school	unit	$f \in \{0.01, 0.05, 0.1, 0.2, 0.5\}$
Contact-high-school	cardinality	$f \in \{0.01, 0.05, 0.1, 0.2, 0.5\}$
Florida Bay food web	submodular	direct k -grid, $k \in \{1, 2, 3, 5, 7, 10, 15\}$
Oil trade	submodular	direct k -grid, $k \in \{3, 5, 10\}$

For Trivago-clicks and Amazon-reviews, we report how sensitive F1 is to the fraction f in their corresponding sections.

D.4 Theory-Experiment bridge for seed injection mass

The theoretical analysis of Section 3 uses the degree-weighted seed injection $\Delta(v) = \delta d_v$ for $v \in S$ and $\Delta(v) = 0$ otherwise. The experiments in Section 4 and Section D.3 use a single-seed rule $\Delta[s^*] = \delta_{\text{exp}} \cdot \text{vol}(\mathcal{T})$ with $s^* \in C$ and target $T = C$ (this is also the case for [Fountoulakis et al., 2021]). The following proposition makes the identification between these two rules precise.

Proposition 3. *Let $S = \{s^*\}$ with $s^* \in C$, $T = C$, and let $\Delta[s^*] = \delta_{\text{exp}} \cdot \text{vol}(\mathcal{T})$ with $\delta_{\text{exp}} \geq 3$. Define the seed-specific*

$$\delta := \frac{\delta_{\text{exp}} \cdot \text{vol}(\mathcal{T})}{d_{s^*}}.$$

Then:

- (a) Δ agrees with the degree-weighted rule $\Delta(v) = \delta d_v$ for $v \in S$.
- (b) Assumption 1 holds with $\beta = 1$ and any $\alpha \leq d_{s^*}/\text{vol}(C)$; in particular, since $\delta = \delta_{\text{exp}} \text{vol}(C)/d_{s^*}$ and $\delta_{\text{exp}} \geq 3$, it holds with the choice $\alpha = 3/\delta$.
- (c) Assumption 3 holds whenever $\sigma \leq \Phi(C)/3$.
- (d) Theorems 1, 2, and 3; Corollaries 1, and 2; and Proposition 1 apply with the seed-specific δ as above. The lower bound $D(x^*) \geq D_C$ (Auxiliary Lemma 3) and the sweep bound retains the same form $3K/\alpha$, now evaluated with the seed-specific $\alpha = 3/\delta$. The box radius and subgradient constant become

$$\bar{x} = \frac{\delta - 1}{\sigma}, \quad B = (\delta - 1)\left(\frac{1}{\sigma} + 2\right),$$

with the seed-specific δ .

Proof. Part (a). Direct: $\delta d_{s^*} = \delta_{\text{exp}} \cdot \text{vol}(\mathcal{T})$ by construction.

Part (b). Since $S = \{s^*\} \subseteq C$, we have $\text{vol}(S \cap C) = d_{s^*} = \text{vol}(S)$, so $\beta = 1$. The tight overlap ratio is

$$\alpha_{\text{tight}} = \frac{\text{vol}(S \cap C)}{\text{vol}(C)} = \frac{d_{s^*}}{\text{vol}(C)}.$$

Since $T = C$, $\text{vol}(\mathcal{T}) = \text{vol}(C)$, and the defining relation $\delta = \delta_{\text{exp}} \text{vol}(\mathcal{T})/d_{s^*}$ gives $\delta = \delta_{\text{exp}} \text{vol}(C)/d_{s^*}$. Hence

$$\frac{3}{\delta} = \frac{3d_{s^*}}{\delta_{\text{exp}} \text{vol}(C)} \leq \frac{d_{s^*}}{\text{vol}(C)} = \alpha_{\text{tight}},$$

because $\delta_{\text{exp}} \geq 3$. Assumption 1 therefore holds with the looser choice $\alpha = 3/\delta \leq \alpha_{\text{tight}}$ and $\beta = 1$.

Part (c). Assumption 3 requires $\delta = 3/\alpha$, $0 \leq w_e(A) \leq 1$, and $\sigma \leq \beta\Phi(C)/3$. The first holds with the choice $\alpha = 3/\delta$ from part (b); the second is inherited from the cut-cost normalization of Section 2; and the third reduces to $\sigma \leq \Phi(C)/3$ since $\beta = 1$.

Part (d). We verify that Auxiliary Lemma 3 goes through unchanged. Using $x = \lambda \mathbf{1}_C$ and $S = \{s^*\} \subseteq C$,

$$\langle \Delta - d, \lambda \mathbf{1}_C \rangle = \lambda(\delta_{\text{exp}} \cdot \text{vol}(\mathcal{T}) \cdot \mathbf{1}_{s^* \in C} - \text{vol}(C)) = \lambda(\delta_{\text{exp}} - 1) \text{vol}(C) \geq 2\lambda \text{vol}(C),$$

using $T = C$ and $\delta_{\text{exp}} \geq 3$. This matches the lower bound used in the proof of Auxiliary Lemma 3, so $D(\lambda \mathbf{1}_C) \geq D_C$ and hence $D(x^*) \geq D_C$. All subsequent arguments (Theorem 2, Corollary 2) are unchanged. The constants $\bar{x}$ and B in Proposition 1 are stated in terms of δ and σ , so they specialize with the seed-specific δ given above. $\square$

Remark on $\delta_{\text{exp}} > 3$. For datasets where $\delta_{\text{exp}} > 3$ (e.g., Amazon with $\delta_{\text{exp}} \in \{50, 200\}$), the seed-specific δ is correspondingly larger. Part (b) still applies with $\alpha = 3/\delta$, which is smaller than α_{tight} ; Assumption 1 holds at this looser α because it is a lower bound. The sweep constant $3K/\alpha$ grows with δ_{exp} , reflecting the over-injection relative to the tight seed–cluster overlap.

Notation convention for experimental injection. In the experimental sections, when we report values such as $\delta = 3$, $\delta = 50$, or $\delta = 200$, this symbol denotes the experimental injection multiplier δ_{exp} used in the single-seed rule $\Delta[s^*] = \delta_{\text{exp}} \text{vol}(T)$, following the convention of [Fountoulakis et al., 2021]. This experimental multiplier should not be confused with the degree-weighted theoretical coefficient δ used in Section 3, where $\Delta(v) = \delta d_v$ for $v \in S$. For a single seed s^* , the two are related by $\delta_{\text{theory}}(s^*) = \frac{\delta_{\text{exp}} \text{vol}(T)}{d_{s^*}}$. Experimental hyperparameter values use the shorthand $\delta \equiv \delta_{\text{exp}}$.

D.5 Trivago-Clicks

Table 5 shows the cluster codes used in the Table 1 for the Trivago dataset. Tables 6 and 7 show the median F1 across the full fraction grid, demonstrating that performance is stable over a range of f values. The cardinality and submodular cut-costs require larger fractions than the unit cut-cost because the weighted in-degree $d_{\text{in}}^{\text{card}}(u)$ is generally smaller than its unit counterpart, making the score more selective; consequently the auto- k rule with a given f activates fewer vertices in the cardinality and submodular settings. Accordingly, we draw f from $[0.01, 0.05]$ for unit cut-cost and from $[0.01, 0.10]$ for cardinality cut-cost.

D.6 Results on Amazon-reviews Dataset

The Amazon-reviews dataset [Veldt et al., 2020, Ni et al., 2019] is a large-scale hypergraph with $n = 2,268,231$ nodes, $m = 4,285,363$ hyperedges, and mean degree $\bar{d} \approx 32.3$. Hyperedges are sets of product reviews, and nodes are labeled by product category. Following Fountoulakis et al. [2021], we evaluate on 9 category clusters with varying injection coefficients ($\delta = 200$ for small clusters, $\delta = 50$ for large clusters). For each cluster $\mathcal{T}$, we run $\min(|\mathcal{T}|, 500)$ single-seed trials: every target-cluster vertex is used once as a seed when $|\mathcal{T}| \leq 500$, and 500 vertices are sampled uniformly at random from $\mathcal{T}$ otherwise. We set $\sigma = 10^{-4}$, $\gamma = 0.5$, and run TL-HFD for $T = 1000$ iterations (unit cut-cost only, matching the baseline). Auto- k fractions are swept over $f \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.03, 0.05, 0.07, 0.1\}$, with fractions producing $k > |\mathcal{T}|$ skipped. For each cluster, we select f by minimum output conductance $\Phi(S_h)$ across the grid (ties are resolved by selecting the F1 corresponding to the smallest such f) — an unsupervised criterion that uses no ground-truth labels — and report the median F1 across trials at the selected f . The sensitivity to the structural-commitment exponent γ is provided in Appendix D.13.

Table 5: Trivago-clicks: Cluster Codes for Table 1. $|\mathcal{T}|$ is the target cluster size; f is the boundary-fraction; k is the resulting per-iteration activation count. γ is set to 1 for both TL-HFD cut costs.

Cluster	$ \mathcal{T} $	TL-UHFD (unit)		TL-CHFD (card.)	
		f	k	f	k
KOR (South Korea)	945	0.02	74	0.07	47
ISL (Iceland)	202	0.03	25	0.07	11
PRI (Puerto Rico)	144	0.02	9	0.03	3
CRM (Crimea)	200	0.01	11	0.03	6
VNM (Vietnam)	832	0.01	23	0.05	21
HKG (Hong Kong)	536	0.01	23	0.02	17
MLT (Malta)	157	0.01	5	0.07	6
GTM (Guatemala)	199	0.01	7	0.10	12
UKR (Ukraine)	264	0.01	6	0.10	12
EST (Estonia)	158	0.01	8	0.05	8

Table 6: Trivago-clicks: median F1 vs. fraction f for unit cut-cost (vol-based k -scaling).

Cluster	$f = .01$	.02	.03	.05
KOR (South Korea)	0.81	0.86	0.81	0.62
ISL (Iceland)	0.98	0.98	0.98	0.96
PRI (Puerto Rico)	0.97	0.97	0.97	0.97
CRM (Crimea)	0.85	0.84	0.84	0.83
VNM (Vietnam)	0.58	0.55	0.50	0.34
HKG (Hong Kong)	0.65	0.49	0.46	0.09
MLT (Malta)	0.97	0.96	0.78	0.79
GTM (Guatemala)	0.95	0.95	0.95	0.95
UKR (Ukraine)	0.65	0.65	0.51	0.38
EST (Estonia)	0.79	0.56	0.56	0.55

Observation: $f \in [0.01, 0.02]$ is broadly robust for Trivago’s unit cut-cost. Performance degrades outside this range, with larger fractions causing overshoot (activating too many boundary vertices per iteration) and smaller fractions causing under-exploration.

Table 7: Trivago-clicks: median F1 vs. fraction f for cardinality cut-cost (n_{est} -based k -scaling).

Cluster	$f = .01$	.02	.03	.05	.07	.10
KOR (South Korea)	0.24	0.75	0.82	0.81	0.82	0.77
ISL (Iceland)	0.07	0.09	0.59	0.95	0.98	0.94
PRI (Puerto Rico)	0.07	0.91	0.97	0.97	0.97	0.97
CRM (Crimea)	0.30	0.71	0.85	0.70	0.66	0.75
VNM (Vietnam)	0.20	0.22	0.30	0.32	0.32	0.32
HKG (Hong Kong)	0.02	0.76	0.73	0.02	0.03	0.01
MLT (Malta)	0.05	0.23	0.96	0.97	0.97	0.97
GTM (Guatemala)	0.05	0.17	0.96	0.96	0.97	0.97
UKR (Ukraine)	0.03	0.08	0.42	0.59	0.61	0.73
EST (Estonia)	0.07	0.08	0.10	0.58	0.56	0.42

Observation: Cardinality requires larger fractions than unit, reflecting the finer-grained weighted d_{in} which makes the scoring more selective. Hong Kong exhibits instability at $f > 0.03$, suggesting the cluster boundary is sharp and sensitive to over-expansion.

Results. Table 8 shows the results. TL-UHFD improves over U-HFD on 6 out of 9 clusters, ties on one, loses on two. For the $\delta = 200$ clusters (small, well-separated categories), TL-UHFD wins on all five. The largest absolute gain is +0.15 F1 on Appliances (0.46 $\rightarrow$ 0.61, a 33% relative improvement), followed by +0.04 on Gift Cards (0.82 $\rightarrow$ 0.86) and +0.04 on Magazine Subscriptions (0.71 $\rightarrow$ 0.75); Fashion and Beauty show modest gains of +0.02 each. For the $\delta = 50$ clusters

Table 8: Amazon-reviews: Best result per cluster (unit costs) in **bold**. Clusters are grouped by injection coefficient δ (smaller clusters use $\delta = 200$; larger use $\delta = 50$).

Method	2	1	3	12	18	15	17	24	25
	Fashion	Beauty	Appliances	Gift Cards	Magazine	Ind./Sci.	Lux. Beauty	Pantry	Software
U-HFD	0.26	0.07	0.46	0.82	0.71	0.02	0.09	0.83	0.09
TL-UHFD	0.28	0.09	0.61	0.86	0.75	0.01	0.09	0.79	0.11
U-LH-2.0	0.21	0.07	0.23	0.29	0.21	0.05	0.06	0.28	0.05
U-LH-1.4	0.21	0.09	0.35	0.40	0.31	0.01	0.07	0.35	0.06
ACL	0.20	0.07	0.22	0.25	0.17	0.04	0.05	0.20	0.04

(larger categories), results are more mixed: TL-UHFD ties U-HFD on Luxury Beauty (both at 0.09) and edges ahead by +0.02 on Software, while losing -0.04 on Prime Pantry, where U-HFD’s primal method finds the cluster effectively. Industrial & Scientific fails for all diffusion-based methods (best F1 = 0.05, achieved by U-LH-2.0), indicating this cluster lacks well-defined local structure amenable to local clustering (baseline conductance > 0.5). TL-UHFD outperforms LH and ACL baselines, often by large margins — e.g., +0.46 over U-LH-1.4 on Magazine Subscriptions and +0.57 over ACL on Gift Cards. Table 9 reports the sensitivity of F1 to the boundary fraction f , confirming that the unsupervised conductance-based selection picks fractions in regions of stable performance.

Table 9: Amazon-reviews: median F1 vs. fraction f for unit cut-cost (vol-based k -scaling). N/A indicates $k > |\mathcal{T}|$ (skipped). * indicates the cluster with lowest conductance that is chosen and reported in Table 8. The Indust. & Sci. row is truncated at two decimal digits.

Cluster	$f = .001$	.002	.005	.01	.02	.03	.05	.07	.10
Amazon Fashion	0.23	0.23	0.28*	0.27	N/A	N/A	N/A	N/A	N/A
All Beauty	0.05	0.08	0.09*	0.09	0.08	N/A	N/A	N/A	N/A
Appliances	0.22	0.22	0.22	0.28	0.28	0.62	0.62	0.61*	0.61
Gift Cards	0.25	0.34	0.53	0.73	0.86*	0.82	0.72	N/A	N/A
Magazine Sub.	0.09	0.20	0.31	0.48	0.75*	0.76	0.65	N/A	N/A
Indust. & Sci.	0.00	0.00	0.00	0.00	0.00	0.01*	0.01	0.01	N/A
Luxury Beauty	0.01	0.03	0.04	0.08	0.08	0.08	0.09*	N/A	N/A
Prime Pantry	0.09	0.20	0.31	0.35	0.79*	0.74	N/A	N/A	N/A
Software	0.02	0.03	0.05	0.09	0.11*	0.12	0.12	N/A	N/A

Observation: $\delta = 200$ clusters peak at $f \in [0.005, 0.03]$. $\delta = 50$ clusters need larger fractions ($f \in [0.02, 0.05]$) to generate sufficient boundary activations. Industrial & Scientific fails across all fractions for both methods, indicating poor cluster separability.

D.7 Florida Bay Food Web

The Florida Bay food web [Li and Milenkovic, 2017] is a small but extremely dense hypergraph with $n = 126$ nodes, $m = 141,233$ hyperedges (all 4-uniform), and mean degree $\bar{d} \approx 4,484$. It contains 3 clusters corresponding to trophic levels: Pelagic ($|\mathcal{T}| = 17$), Detritus ($|\mathcal{T}| = 35$), and Benthic ($|\mathcal{T}| = 70$). Each hyperedge encodes a motif between two prey and two predator species.

Why submodular cut-cost only. Following Fountoulakis et al. [2021], we evaluate using the *submodular* cut-cost (S-HFD), which is the natural model for this dataset. The submodular weight function w_e distinguishes between prey–prey splits ($w_e = 0$), predator–predator splits ($w_e = 0$), and prey–predator splits ($w_e = 1$). Unit and cardinality cut-costs treat all splits equally, losing this biological distinction. The HFD paper uses the food web specifically to demonstrate the advantage of submodular cut-costs over simpler alternatives.

Setup. We set $\sigma = 10^{-4}$, $T = 500$ iterations, and we use every target-cluster vertex once as a seed (matching the HFD protocol). The injection coefficients are cluster-specific: $\delta_{exp} = 20$ (Pelagic),

$\delta_{exp} = 10$ (Detritus), $\delta_{exp} = 5$ (Benthic). We compare S-HFD and our method (TL-HFD) with top- k , $\gamma \in \{0.25, 0.5, 1.0\}$ over $k \in \{1, 2, 3, 5, 7, 10, 15\}$.

Unlike the other datasets, we sweep over a grid of k values rather than using the fraction-based k selection. The food web’s extreme density ($\bar{d} \approx 4,484$ with only 126 nodes) makes the volume-based scaling $k = f \cdot \text{vol}_{\text{est}}$ produce unreasonably large k values — for instance, Detritus has $\text{vol}_{\text{est}} = 173,311$, so even $f = 0.001$ yields $k = 173$, exceeding the total number of nodes. The auto- k mechanism is designed for sparse-to-moderate hypergraphs where volume scales proportionally with cluster size; it is not applicable in this ultra-dense regime.

The scoring for submodular cut-cost uses the submodular weight function to compute d_{in} : for each edge e incident to boundary vertex u that touches the active set A , $d_{\text{in}}(u) += w_e(A \cap e)$.

Table 10: Florida Bay food web: median F1 for submodular cut-cost. Best result per cluster in **bold**.

Cluster	$ \mathcal{T} $	S-HFD	TL-HFD	(γ, k)
Pelagic	17	0.69	0.69 ($k \leq 15$)	any
Detritus	35	0.62	0.68 ($k = 3$)	$\gamma = 0.25$
Benthic	70	0.84	0.84 ($k = 3$)	$\gamma = 0.25$

Results. Table 10 summarizes the results. The food web is a dense dataset: it has only 126 nodes but 141,233 hyperedges, giving a mean degree of approximately 4,484. Thus, each node participates in thousands of hyperedges. As a consequence, the boundary at each iteration is often smaller than the tested values of k , so thresholding has limited effect: in many runs, all boundary vertices are activated.

For Pelagic, TL-SHFD matches S-HFD, achieving median F1 = 0.69 across almost all tested k and γ values. For Detritus, TL-HFD improves over S-HFD, increasing median F1 from 0.62 to 0.68 at $\gamma = 0.25$ and $k = 3$. For Benthic, S-HFD and TL-SHFD attain the same result, achieving F1 = 0.84, at $\gamma = 0.25$ and $k = 3$.

These results show that TL-HFD with submodular cut-cost correctly integrates the submodular weight function in both the subgradient computation and the boundary scoring, while largely preserving the HFD’s clustering quality on this dense, well-structured dataset.

D.8 Contact-High-School

The contact-high-school dataset [Mastrandrea et al., 2015] is a dense hypergraph of face-to-face interactions among students, with $n = 327$ nodes, $m = 7,818$ hyperedges, and mean degree $\bar{d} \approx 55$. The nine ground-truth clusters correspond to school classes, such as 2BIO1, MP*1, and PC. Following Fountoulakis et al. [2021], we evaluate each class by using every target-cluster vertex once as a single seed, yielding $|\mathcal{T}|$ single-seed trials per class.

Setup. We set $\sigma = 10^{-4}$, injection factor $\delta_{\text{exp}} = 3$, and exponent $\gamma = 1.0$. TL-HFD is run for $T = 1000$ iterations for both unit and cardinality cut-costs. For each cluster, f is selected from the grid by minimum output conductance, and the median F1 at the selected f is reported.

Results. Tables 11 and 12 show the results. This dataset is a challenging evaluation setting because HFD (Fountoulakis et al. [2021]) already achieves near-perfect F1 on most classes, leaving limited room for improvement.

For unit cut-cost, TL-UHFD wins on three classes: 2BIO3 (+0.027), MP*2 (+0.081), and PC (+0.121). It ties HFD on four classes and loses on two classes, 2BIO2 (−0.036) and MP*1 (−0.071), where HFD is already perfect or near-perfect.

For cardinality cut-cost, TL-CHFD ties or wins on eight out of nine classes. MP*1 improves from 0.964 to 1.000, and MP*2 improves from 0.796 to 0.833. The only loss is 2BIO2 (−0.054), where HFD achieves perfect F1. TL-CHFD achieves F1 = 1.0 on five classes: 2BIO3, MP*1, PSI*, PC, and PC*. Overall, these results show that TL-HFD largely preserves quality on a dense, well-clustered hypergraph where the baseline is already strong, while still providing gains on some classes.

Table 11: Contact-high-school: median F1 for unit cut-cost. f is selected per cluster by minimum output conductance for TL-UHFD. Best result per cluster in **bold**.

Cluster	$ \mathcal{T} $	U-HFD	TL-UHFD	f
2BIO1	36	0.986	0.986	0.01
2BIO2	34	1.000	0.964	0.05
2BIO3	40	0.593	0.620	0.01
MP*1	29	0.964	0.893	0.01
MP*2	38	0.731	0.812	0.50
PSI*	34	1.000	1.000	0.01
PC	44	0.879	1.000	0.01
PC*	39	1.000	1.000	0.01
MP	33	0.985	0.985	0.01

Table 12: Contact-high-school: median F1 for cardinality cut-cost. Best result per cluster in **bold**.

Cluster	$ \mathcal{T} $	C-HFD	TL-CHFD	f
2BIO1	37	0.986	0.986	0.01
2BIO2	34	1.000	0.946	0.01
2BIO3	40	1.000	1.000	0.01
MP*1	29	0.964	1.000	0.01
MP*2	38	0.796	0.833	0.01
PSI*	34	1.000	1.000	0.01
PC	44	1.000	1.000	0.01
PC*	39	1.000	1.000	0.01
MP	33	0.985	0.985	0.01

D.9 Oil Trade Ranking

The oil-trade dataset [Fountoulakis et al., 2021] is a 4-uniform hypergraph of international oil-trade flows with $n = 229$ countries, $m = 100,639$ hyperedges, and mean degree $\bar{d} \approx 1,758$. Following Fountoulakis et al. [2021], we use this dataset for a node-ranking task rather than a clustering task: given a seed country, Iran, the goal is to rank other countries by proximity in the diffusion embedding. There are no ground-truth cluster labels, so evaluation is qualitative.

Setup. We set $\Delta[\text{Iran}] = 500,000$, $\sigma = 0.1$, and run the submodular cut-cost with $T = 20$ iterations, matching the HFD parameter setting. We also run unit and cardinality cut costs for $T = 100$ to observe the ranking generated from TL-HFD. For TL-SHFD (TL-HFD with submodular cut costs), rankings are computed from the best dual iterate: $\text{score}(v) = \sigma \cdot x_v^{\text{dual}}$. TL-HFD uses a grid over k with $k \in \{3, 5, 10\}$ and $\gamma \in \{0.25, 0.5, 1.0\}$.

Why submodular is the correct model. The oil-trade hypergraph encodes 4-way trade motifs in which two countries export and two countries import within each transaction. The submodular cut-cost distinguishes between exporter–exporter splits, which receive low penalty, and exporter–importer splits, which receive high penalty. This captures the directional structure of trade flows. Unit and cardinality cut-costs treat all nontrivial splits symmetrically, and therefore discard this exporter–importer structure.

Results. Table 13 compares the submodular rankings. Under the submodular cut-cost, TL-HFD produces identical rankings across all tested k and γ configurations, indicating that thresholding does not materially change the ranking on this dense instance. The top-10 ranking overlaps with the HFD baseline on 9 of 10 countries, with only minor reordering: Guinea moves from rank 5 to rank 3, and Oman drops out in favor of Mozambique. Both methods identify countries with oil-trade network positions similar to Iran, such as Turkmenistan, Iraq, Bahrain, Kazakhstan, Libya, and Qatar.

Under unit and cardinality cut-costs, the corresponding HFD and TL-HFD variants rank countries near Iran that are less clearly related to Iran’s oil-trade position, such as Kenya, Bangladesh, Arme-

Table 13: Oil-trade ranking (seed: Iran): top-10 countries under submodular cut-cost. TL-HFD produces identical rankings across all k and γ configurations.

Rank	S-HFD (baseline)	TL-SHFD (all configs)
1	Turkmenistan	Turkmenistan
2	Iraq	Iraq
3	Bahrain	Guinea $\uparrow$
4	Brunei	Bahrain $\downarrow$
5	Guinea	Brunei $\downarrow$
6	Kazakhstan	Kazakhstan
7	Libya	Libya
8	Oman	Qatar $\uparrow$
9	Qatar	Congo $\uparrow$
10	Congo	Mozambique (new)

9 of 10 countries overlap. Oman (baseline rank 8) is replaced by Mozambique. All are oil-producing or oil-exporting nations with similar trade network positions to Iran.

nia, and Tanzania. This behavior is consistent across methods and reflects model mismatch: unit and cardinality cut-costs cannot encode the directional exporter–importer structure of the 4-way trade motifs. The submodular cut-cost is therefore necessary for meaningful rankings on this dataset, as also observed by Fountoulakis et al. [2021].

D.10 Hypergraph Stochastic Block Model (SBM)

We evaluate TL-HFD on synthetic hypergraphs generated from the hypergraph stochastic block model used by Fountoulakis et al. [2021]. The hypergraph has $n = 100$ vertices partitioned into two equal blocks of size 50, and all hyperedges are 3-uniform. We use the first block as the target cluster $\mathcal{T}$. The intra-block triples are sampled with probability $p = 0.0765$, which gives approximately 1500 intra-block hyperedges in expectation for each block. The inter-block probability q is varied to control the ground-truth conductance $\Phi(\mathcal{T})$ of the target cluster.

Conductance sweep. The sweep over q corresponds to approximately 500 to 9000 expected cross-block hyperedges, and hence to total expected hyperedge counts ≈ 3500 to 12000. This varies the target-cluster conductance from approximately 0.10 to 0.50. For each setting of q , one hypergraph is sampled, and every vertex in the target cluster is used once as the single seed, yielding 50 single-seed trials per hypergraph. This keeps the number of vertices, hyperedge cardinality, and intra-block density fixed while varying the amount of cross-block noise and hence the separation of the planted cluster.

Methods compared. We compare U-HFD against TL-UHFD with fixed top- k : activate the top k boundary vertices ranked by the score $s(u) = \kappa(u) (d_{\text{in}}(u)/d(u))^\gamma$, with $k \in \{3, 5, 10\}$ and $\gamma \in \{0.3, 0.5, 1\}$. Because cluster size is fixed at $|\mathcal{T}| = 50$ and the boundary is bounded by $|V \setminus \mathcal{T}| \leq 50$, we sweep k directly rather than via the auto- k rule.

Hypotheses. The synthetic SBM tests two hypotheses about TL-HFD’s behavior:

1. **H1 (Effect of γ).** The planted partition model has a clean, symmetric block structure. Unlike real-world hypergraphs, it contains no semantically ambiguous bridge vertices — such as products belonging naturally to multiple categories — so we expect the structural commitment factor $(d_{\text{in}}/d)^\gamma$ to have only a minor effect, with most of the variation between methods driven by the top- k activation cap itself. SBM thereby provides a controlled setting for isolating the effect of thresholding from the effect of structural commitment.
2. **H2 (Effect of conductance).** We expect TL-HFD to recover the planted cluster well when conductance is low (clean separation, little inter-cluster noise) and to degrade as conductance increases. Specifically, restricting boundary expansion may slow recovery on low-conductance clusters where broad expansion is beneficial, but the same cap may help on high-conductance clusters by limiting over-inclusion of non-target vertices. The

crossover — if it exists — locates the conductance regime in which top- k thresholding becomes net beneficial.

Setup. We use $\sigma = 0.01$ and injection factor $\delta_{\text{exp}} = 3$. The choice $\sigma = 0.01$ matches the baseline’s synthetic protocol and is larger than the $\sigma = 10^{-4}$ used on real-world hypergraphs, which have noisier and more heterogeneous local structure. Both U-HFD and TL-HFD are run for $T = 500$ iterations; U-HFD converges in fewer than 100 iterations under its alternating-minimization update. We report median F1 across the 50 single-seed trials per hypergraph.

Results. We report results at low conductance ($\Phi(T) \in [0.10, 0.20]$, Table 14) and at high conductance ($\Phi(T) \in [0.40, 0.50]$, Table 15). In the intermediate range $\Phi(T) \in [0.20, 0.40]$, TL-UHFD with top- k tracks U-HFD closely (within ± 0.02 median F1), so we omit these rows to keep the focus on the regimes where the methods diverge.

Hypothesis 1: Effect of γ . Across the SBM sweep, varying $\gamma \in \{0.3, 0.5, 1.0\}$ produces no meaningful change in F1 scores (differences within ± 0.005 at all conductance levels). This is consistent with the Hypothesis 1: in a planted partition without bridge vertices, all boundary vertices have similar d_{in}/d ratios, so the structural commitment factor rescales the score uniformly and does not change the top- k ranking. We therefore report all SBM results at $\gamma = 1$.

Table 14: Synthetic SBM: median F1 at *low conductance* (clean, well-separated clusters). The columns labeled topk= k give committed top- k at fixed activation count k . U-HFD achieves perfect F1; TL-UHFD remains very close, with a small loss only near the higher end of this low-conductance range.

#edges	$\Phi(T)$	U-HFD	topk=3	topk=5	topk=10
3500	0.095	1.000	1.00	1.00	1.00
3600	0.111	1.000	1.00	1.00	1.00
3700	0.126	1.000	1.00	1.00	1.00
3800	0.141	1.000	1.00	1.00	1.00
3900	0.154	1.000	1.00	1.00	1.00
4000	0.167	1.000	0.99	0.99	0.99
4100	0.179	1.000	0.99	0.99	0.99
4200	0.191	1.000	0.99	0.99	0.99

On well-separated clusters, U-HFD achieves perfect F1. TL-HFD matches within ≤ 0.015 across all configurations, confirming that thresholding causes only a small loss when clusters are easy to find.

Table 15: Synthetic SBM: median F1 at *high conductance* (noisy, poorly separated clusters). The columns labeled topk= k give committed top- k at fixed activation count k . Best per row in **bold**. TL-HFD improves over U-HFD in the intermediate high-conductance regime.

#edges	$\Phi(T)$	U-HFD	topk=3	topk=5	topk=10
7700	0.407	0.74	0.75	0.75	0.74
8000	0.417	0.72	0.73	0.74	0.73
8500	0.432	0.68	0.69	0.69	0.69
9000	0.445	0.64	0.66	0.68	0.67
10000	0.468	0.59	0.60	0.60	0.60
11000	0.486	0.56	0.55	0.55	0.56
12000	0.502	0.51	0.50	0.51	0.50

TL-HFD improves over U-HFD in the intermediate range $\Phi(T) \approx 0.42$ – 0.47 , with the largest rounded gain at $\Phi(T) = 0.445$, where U-HFD obtains 0.64 and top- $k = 5$ obtains 0.68. Beyond $\Phi(T) \approx 0.49$, all methods degrade as the planted cluster becomes weakly separated.

Low conductance. For $\Phi(T) \leq 0.20$, U-HFD achieves perfect F1 across all tested conductance levels. TL-HFD remains very close, with median F1 between 0.99 and 1.00. This is the cost of

thresholding in a clean planted partition: when most positively pushed boundary vertices are useful, restricting expansion can only slow or slightly limit recovery.

Higher conductance. For $\Phi(T) \in [0.40, 0.50]$, the planted cluster is less well separated. In the intermediate range $\Phi(T) \approx 0.42\text{--}0.47$, TL-HFD improves over U-HFD, with the largest rounded gain at 9000 edges ($\Phi(T) = 0.445$): U-HFD obtains 0.64, while TL-HFD with $\text{top-}k = 5$ obtains 0.68. Beyond $\Phi(T) \approx 0.49$, all methods degrade together, reflecting the weak separation of the planted cluster.

Hypothesis 2: Effect of conductance. The two regimes in Tables 14 and 15 confirm H2 and locate the crossover. On low-conductance clusters ($\Phi(T) \leq 0.20$), U-HFD achieves perfect F1 across all tested levels, and TL-HFD remains very close (median F1 between 0.99 and 1.00, gap ≤ 0.015). This is the predicted cost of thresholding in a clean planted partition: when most positively pushed boundary vertices belong to the target, restricting expansion can only slow or slightly limit recovery. On high-conductance clusters ($\Phi(T) \in [0.40, 0.50]$), the planted cluster is less well separated, and the relative ordering between U-HFD and TL-HFD flips in the intermediate range $\Phi(T) \approx 0.42\text{--}0.47$: TL-HFD improves over U-HFD, with the largest rounded gain at $\Phi(T) = 0.445$, where U-HFD obtains 0.64 and TL-HFD with $\text{top-}k = 5$ obtains 0.68. Beyond $\Phi(T) \approx 0.49$, all methods degrade together as the planted structure becomes too weak for any local method to recover. The crossover near $\Phi(T) \approx 0.40$ therefore locates the regime in which $\text{top-}k$ thresholding becomes net beneficial: TL-HFD’s restriction protects against over-inclusion of cross-block vertices once the boundary expansion itself becomes noisy.

Tradeoff summary. The synthetic experiment isolates the role of $\text{top-}k$ thresholding under a clean planted-partition model. On well-separated clusters, thresholding slightly hurts because broad expansion is beneficial. On noisier clusters, thresholding can help by limiting over-expansion. The larger gains observed on real-world datasets are consistent with the additional role of committed weighting, which is designed to filter structurally ambiguous boundary vertices; this effect is largely absent in the planted SBM.

Visualizing the SBM sweep. Figures 1–4 present the SBM results graphically, complementing the median values in the tables. Each figure plots a metric (F1 or output conductance) against the ground-truth target-cluster conductance $\Phi(T)$. Solid and dashed lines show the median across the 50 single-seed trials per hypergraph, with line style and color identifying the method (U-HFD or TL-HFD at a specific $\text{top-}k$ cap). The shaded band around each line shows the interquartile range (25th–75th percentile) across the same trials: a narrow band indicates insensitivity to seed choice on that hypergraph, while a wide band indicates substantial seed-to-seed variability. We report each metric across two panels: a low- Φ panel covering $\Phi(T) \in [0.10, 0.20]$ (clean, well-separated clusters) and a high- Φ panel covering $\Phi(T) \in [0.40, 0.50]$ (noisy, weakly separated clusters). The intermediate range $\Phi(T) \in [0.20, 0.40]$ is omitted because all methods track each other within ± 0.02 median F1 there. For F1, higher is better; for output conductance, lower is better. The methods are visually distinguished by both color and dash pattern, so the figures remain legible in grayscale. Figures 1, and 2 show the F1 scores for U-HFD and TL-HFD under low conductance setting. Figures 3, and 4 depicts how the methods track the ground truth conductance.

D.11 Experiments on h-ABCD Synthetic Datasets

Synthetic data generation. We use the h-ABCD benchmark [Kamiński et al., 2023], the hypergraph extension of the ABCD random graph model [Kamiński et al., 2021]. We generate hypergraphs on $n = 5000$ nodes partitioned into communities whose sizes follow a $(1.9, 100, 500)$ truncated power-law distribution (TPD) with exponent 1.9 and range $[100, 500]$. Each node has a degree drawn from a $(2.2, 3, 50)$ -TPD, and hyperedge sizes follow a $(2.0, 3, 50)$ -TPD, so edges range from small triangles to large group interactions.

We generate two kinds of hyperedges. *Community edges* draw their members primarily from a single community; *background edges* draw members uniformly from the full vertex set. The fraction of each node’s degree assigned to background edges is the *mixing parameter* $\xi \in [0, 1]$: at $\xi = 0$ all hyperedges are internal and community structure is perfect; at $\xi = 1$ all edges are background and the structure is erased.

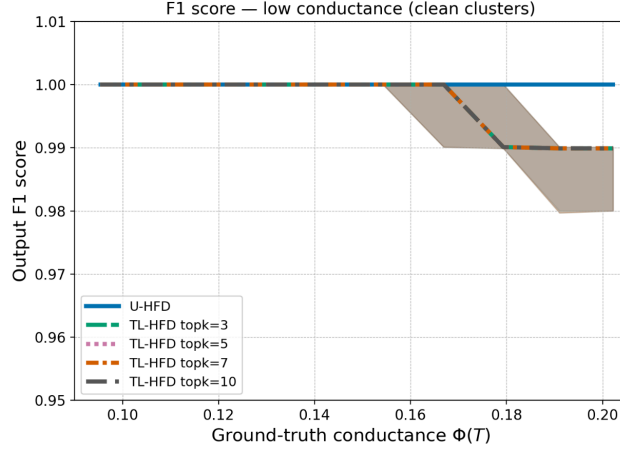


Figure 1: Synthetic SBM: F1 score at low conductance ($\Phi(T) \in [0.10, 0.20]$, easy clusters). U-HFD achieves $F1 = 1.0$; TL-HFD with $k \in \{3, 5, 7, 10\}$ matches within ± 0.015 , indicating that top- k thresholding incurs only a small cost when clusters are clean.

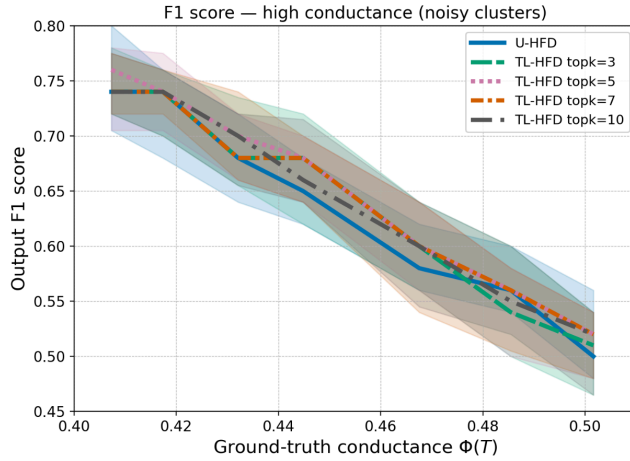


Figure 2: Synthetic SBM: F1 score at high conductance ($\Phi(T) \in [0.40, 0.50]$, noisy clusters). TL-HFD overtakes U-HFD for $\Phi(T) \in [0.42, 0.47]$, with the largest gain at $\Phi(T) = 0.445$. Beyond $\Phi(T) \approx 0.49$, all methods degrade together as the planted structure weakens.

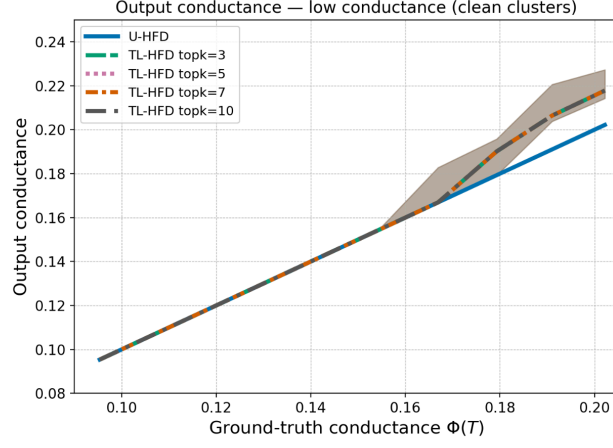


Figure 3: Synthetic SBM: output conductance at low ground-truth conductance ($\Phi(T) \in [0.10, 0.20]$). All methods track the ground-truth conductance closely.

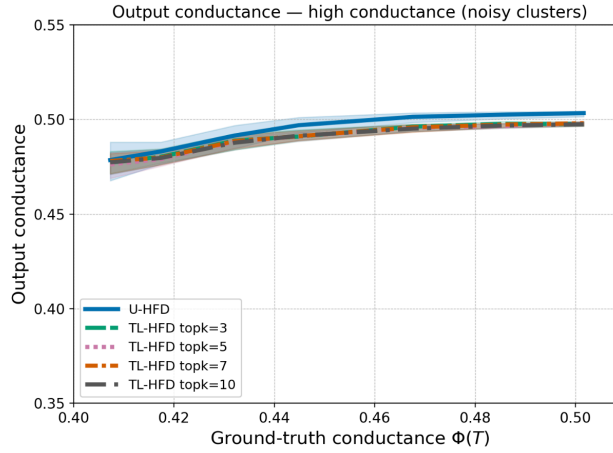


Figure 4: Synthetic SBM: output conductance at high ground-truth conductance ($\Phi(T) \in [0.40, 0.50]$). TL-HFD produces slightly lower (better) conductance clusters than U-HFD in this regime.

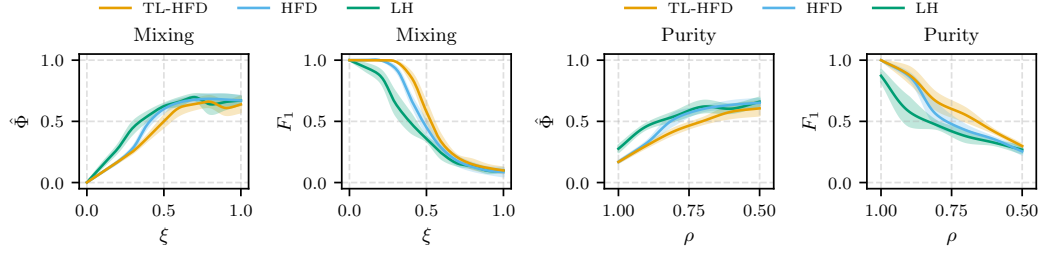


Figure 5: h-ABCD synthetic benchmark: TL-HFD vs. HFD and LH across two robustness sweeps. **Top row (mixing):** output conductance $\hat{\Phi}$ and F1 as a function of the mixing parameter ξ with purity $\rho = 1$. At $\xi = 0$ all hyperedges are internal to a single community; at $\xi = 1$ all hyperedges are drawn uniformly from the full vertex set. **Bottom row (purity):** $\hat{\Phi}$ and F1 as a function of ρ with mixing $\xi = 0.2$. At $\rho = 1$ each community edge is a clique within one community; as $\rho \rightarrow 0.5$, community edges contain only a bare majority of same-community nodes. Each panel shows medians across communities and 10 independently generated hypergraphs, with shaded regions indicating confidence bands across the same trials. TL-HFD maintains higher F1 in the intermediate-mixing regime (top right) and shows a widening advantage as community edges become more heterogeneous (bottom right), consistent with the structural-commitment scoring’s de-prioritization of boundary vertices that span communities.

The *purity* parameter $\rho \in [0.5, 1]$ controls the homogeneity of community edges. A community edge of size d has its number of within-community members drawn from $\text{Binomial}(d, \rho)$, restricted to a strict majority $c > d/2$. At $\rho = 1$ each community edge is a clique within one community; as $\rho \rightarrow 0.5$ edges become increasingly mixed, containing only a bare majority of same-community nodes.

Protocol. For each community we randomly select one of its members as the seed. Hyperparameters for each method are tuned per community via grid search to minimize output conductance — an unsupervised criterion matching the protocol used for real-world datasets. All reported values are medians and confidence bands over communities across 10 trials.

Robustness to cluster strength (Mixing). We vary ξ uniformly over 11 values in $[0, 1]$ while holding $\rho = 1$. This sweep tests whether each method can recover a planted community as the surrounding noise level increases: at low ξ the target is well-separated from the background, while at high ξ community structure approaches uniform random. Figure 5 shows F1 and output conductance across the sweep. All methods recover the planted community in the low-noise regime; the methods diverge in the intermediate regime, where TL-HFD’s structural-commitment scoring filters boundary vertices whose incidences are dominated by background edges. In the high-noise regime, no local method can recover the community, and all methods degrade together.

Robustness to hyperedge heterogeneity (Purity). We fix $\xi = 0.2$ and vary ρ uniformly in $[0.5, 1.0]$. This sweep isolates within-edge heterogeneity from cross-community noise: the total background-edge mass is held fixed, but community edges become progressively more mixed as ρ decreases. Figure 5 shows F1 and output conductance across the sweep. The decrease in ρ exposes genuine bridge vertices — nodes that participate heavily in mixed-membership hyperedges — which are precisely the boundary vertices that TL-HFD’s $(d_{\text{in}}^{(t)}/d)^\gamma$ weighting (Algorithm 1, line 10) is designed to de-prioritize. The widening gap between TL-HFD and the baselines as $\rho \rightarrow 0.5$ is consistent with this mechanism: thresholded selection avoids activating bridge vertices that other methods would include, leading to lower output conductance and higher F1 on the recovered cluster.

D.12 Matched-Convergence and Support-Locality on Amazon-reviews

In this section we compare the computational behavior of TL-HFD and HFD on the Amazon-reviews dataset using two diagnostics. First, we ask how many TL-HFD iterations are needed before its returned cluster matches the final F1 score of HFD. Second, we compare the size of the nonzero

state maintained by each method during the run. The latter should be interpreted as a support-locality measure, not as a complete per-iteration runtime bound: TL-HFD also scores the current boundary $\partial A^{(t)}$, so the full scanned region is $L_t = A^{(t)} \cup \partial A^{(t)}$.

Throughout this section, the active set at iteration t denotes the set of vertices on which the method stores nonzero state:

- For HFD,

$$A_t^{\text{HFD}} = \{v \in \mathcal{V} : e_t(v) > 0\},$$

where $e_t(v)$ is the excess flow at vertex v after the t -th alternating-minimization step.

- For TL-HFD,

$$A_t^{\text{TL}} = \{v \in \mathcal{V} : x_t(v) > 0\},$$

where x_t is the projected-subgradient iterate.

We also track the returned cluster at iteration t , defined for both methods as the sweep-cut set with minimum conductance observed up to iteration t . This is the cluster the method would output if stopped at iteration t .

All results in this section use the nine Amazon categories from Section D.6. Each category is evaluated using 10 random single-seed trials drawn from the same $\min(|\mathcal{T}|, 500)$ -seed pool used in Table 8. The smaller sample is sufficient for illustrating per-iteration convergence behavior; precise F1 estimates are reported in Table 8. All other settings match the headline protocol: TL-HFD uses the conductance-selected fraction f for each category from Table 9, $\sigma = 10^{-4}$, and runs for 1000 iterations; HFD is run for 200 iterations (HFD typically converged in fewer than 200 iterations). We report medians across seeds.

D.12.1 Matched-Convergence Analysis

A direct comparison of total wall-clock time is not the most informative comparison, because HFD uses a primal-dual alternating-minimization routine that typically converges in few iterations, whereas TL-HFD uses projected subgradient steps and is run for many more iterations. We therefore compare the time required for TL-HFD to reach the final F1 score of HFD.

For each seed and category, let F1^{HFD} be the F1 score of HFD’s final returned cluster after 200 iterations. Let F1_t^{TL} be the F1 score of TL-HFD’s returned cluster when stopped at iteration t . We define

$$t^* = \min\{t : \text{F1}_t^{\text{TL}} \geq \text{F1}^{\text{HFD}}\},$$

when such an iteration exists. The projected wall-clock time at t^* is estimated by linearly scaling the measured 1000-iteration TL-HFD runtime:

$$\text{time}_{\text{proj}} = \frac{t^*}{1000} \cdot \text{time}_{\text{TL},1000}.$$

This projection is a coarse estimate. The activated set grows monotonically, but the per-iteration cost is governed by the scanned region $L_t = A^{(t)} \cup \partial A^{(t)}$, which can fluctuate; we therefore treat the projection as an approximate diagnostic.

Table 16 reports HFD’s final F1 and wall-clock time, TL-HFD’s final F1 after 1000 iterations, the median t^* over that matched seeds, and the projected time at t^* .

Findings. On six of the nine categories—Amazon Fashion, Appliances, Gift Cards, Magazine Subscriptions, Luxury Beauty, and Software—TL-HFD reaches HFD’s final F1 on a majority of seeds, with median $t^* \leq 59$ among matched seeds. The projected wall-clock time is often one to two orders of magnitude smaller than HFD’s runtime. For example, on Gift Cards, TL-HFD matches HFD’s final F1 by iteration 7, corresponding to approximately 0.11 seconds of projected wall-clock time, compared with 5.57 seconds for HFD. Continuing TL-HFD beyond this point further improves the median F1 on this category from 0.811 to 0.887. The remaining categories illustrate limitations of the top- k restriction. Industrial & Scientific is difficult for both methods (HFD F1 = 0.010), and TL-HFD performance is not competitive on this category (F1 = 0.008), suggesting the cluster lacks local structure that either method can recover.

Table 16: Matched-convergence analysis on Amazon-reviews, computed over 10 single-seed trials per cluster (subsample of the $\min(|\mathcal{T}|, 500)$ -seed protocol used in Table 8). t^* is the median first iteration at which TL-HFD’s returned cluster reaches HFD’s final F1, taken over seeds where this occurs within 1000 iterations; $t^* = \text{—}$ indicates TL-HFD never reached HFD’s F1 on any seed. Projected time scales TL-HFD’s 1000-iteration wall-clock by $t^*/1000$.

Cluster	δ	HFD F1	HFD time (s)	TL F1	t^*	Projected (s)
Amazon Fashion	200	0.145	5.81	0.278	2	0.01
All Beauty	200	0.061	7.16	0.087	7	0.12
Appliances	200	0.476	1.09	0.648	10	0.01
Gift Cards	200	0.811	5.57	0.887	7	0.11
Magazine Subscriptions	200	0.710	6.09	0.777	11	0.10
Industrial & Scientific	50	0.010	23.15	0.008	—	—
Luxury Beauty	50	0.046	12.79	0.132	10	0.91
Prime Pantry	50	0.799	31.00	0.755	7	1.32
Software	50	0.077	8.11	0.269	59	1.67

D.12.2 Active-Set Growth

We next compare the size of the nonzero state maintained by each method during the run. For HFD, Fountoulakis et al. [2021] provide a sparsity guarantee for the final active set, but this does not directly control the size of intermediate active sets. In contrast, since all Amazon trials are single-seed runs, TL-HFD activates at most k new boundary vertices per iteration, so

$$|A_t^{\text{TL}}| \leq 1 + kt,$$

with equality only if every activated coordinate remains positive.

Table 17 compares HFD’s peak and final active-set sizes with TL-HFD’s final support size, reporting both absolute counts and ratios to the target-cluster size $|\mathcal{T}|$.

Table 17: Active-set sizes across HFD and TL-HFD iterations on Amazon-reviews, reported as medians across the 10 seeds used in this sub-experiment (cf. Table 16). Parentheses give the ratio to the target cluster size $|\mathcal{T}|$. $|A_{\max}^{\text{HFD}}|$ is the largest HFD active set during the run, $|A_T^{\text{HFD}}|$ is HFD’s final active set, and $|A_T^{\text{TL}}|$ is TL-HFD’s final support size. The column k reports the per-iteration activation cap used by TL-HFD on each category.

Cluster	$ \mathcal{T} $	$ A_{\max}^{\text{HFD}} $	$ A_T^{\text{HFD}} $	$ A_T^{\text{TL}} $	k
Amazon Fashion	31	3596 (116 $\times$)	3408 (110 $\times$)	24 (0.77 $\times$)	15
All Beauty	85	4519 (53 $\times$)	3444 (41 $\times$)	5 (0.06 $\times$)	4
Appliances	48	202 (4.2 $\times$)	195 (4.1 $\times$)	26 (0.54 $\times$)	5
Gift Cards	148	2870 (19 $\times$)	1992 (13 $\times$)	106 (0.72 $\times$)	59
Magazine Subscriptions	157	2274 (14 $\times$)	1456 (9.3 $\times$)	121 (0.77 $\times$)	69
Industrial & Scientific	5334	14354 (2.7 $\times$)	12504 (2.3 $\times$)	73 (0.01 $\times$)	72
Luxury Beauty	1581	7313 (4.6 $\times$)	7083 (4.5 $\times$)	2024 (1.28 $\times$)	1404
Prime Pantry	4970	33759 (6.8 $\times$)	29405 (5.9 $\times$)	4980 (1.00 $\times$)	2622
Software	802	4350 (5.4 $\times$)	4211 (5.3 $\times$)	371 (0.46 $\times$)	356

Findings. HFD’s active set often expands far beyond the target cluster size during its intermediate iterations. This effect is most pronounced on small target categories: Amazon Fashion peaks at $116 \times |\mathcal{T}|$, All Beauty at $53 \times |\mathcal{T}|$, and Gift Cards at $19 \times |\mathcal{T}|$. HFD’s active set can then shrink between its peak and final iteration as excess is redistributed, but the final HFD active set remains larger than the target cluster on all nine categories.

TL-HFD maintains substantially smaller nonzero supports. On seven of nine categories, the final TL-HFD support is strictly below the target size. The two exceptions are Luxury Beauty, where the final support is $1.28 \times |\mathcal{T}|$, and Prime Pantry, where it is essentially equal to the target size (4980 versus 4970 vertices). These numbers should be interpreted as support-locality measurements rather than full scanned-work measurements, because TL-HFD still scores the current boundary at each iteration.

Discussion. The two diagnostics tell a consistent story. The matched-convergence analysis shows that TL-HFD often reaches HFD-level F1 well before the end of its 1000-iteration run. The active-set analysis shows that it does so while maintaining a much smaller nonzero state than HFD. On most categories, the final TL-HFD support is below the target-cluster size; the exceptions are Luxury Beauty (support $1.28 \times |\mathcal{T}|$) and Prime Pantry (support $\approx |\mathcal{T}|$).

These results highlight the tradeoff between the two solvers. HFD converges in fewer iterations but can maintain large transient active sets. TL-HFD uses a slower first-order method, but its top- k boundary activation directly controls the growth of the maintained support. This support-locality property does not by itself imply a full worst-case runtime bound, since each TL-HFD iteration still scores the current boundary. Together, however, these diagnostics indicate that on the Amazon categories where TL-HFD succeeds, its top- k schedule achieves fast F1 progress while keeping the nonzero support small.

D.13 Sensitivity to the structural-commitment exponent γ

The boundary score in Algorithm 1 reweights the gradient push $\kappa(u)$ by a structural-commitment factor:

$$s^{(t)}(u) = \kappa^{(t)}(u) \left(\frac{d_{\text{in}}^{(t)}(u)}{d_u} \right)^\gamma,$$

where the exponent γ controls how strongly the score favors boundary vertices that are well attached to the current active region. The Amazon-reviews results in Section D.6 use $\gamma = 0.5$ as it provides the lowest conductance. This appendix studies how the cluster-recovery quality of TL-HFD varies with γ .

Setup. We sweep $\gamma \in \{0.25, 0.50, 0.75, 1.00\}$ on the nine Amazon-reviews clusters of Section 4. Each ablation run uses $n = 50$ random target seeds per cluster and the same fraction grid as Table 9. All other hyperparameters (σ , δ_{exp} , iteration budget, cut-cost) are unchanged. For each cluster and each γ , we report the F1 and output conductance at the fraction f selected by minimum output conductance over the sweep, with ties broken by the smallest f . We analyze sensitivity in relative terms with respect to $\gamma = 0.5$ (for the ablation run), defining

$$\Delta_{F1}(\gamma, c) = \frac{F1(\gamma, c) - F1(0.5, c)}{F1(0.5, c)}, \quad \Delta_\Phi(\gamma, c) = \frac{\Phi(\gamma, c) - \Phi(0.5, c)}{\Phi(0.5, c)}.$$

Relative F1 change. Table 18 reports Δ_{F1} relative to $\gamma = 0.5$. Industrial & Scientific has $F1(\gamma) < 0.04$ at every γ ; we mark its row degenerate, since relative changes amplify near-zero F1 values without conveying meaningful sensitivity.

Table 18: Amazon-reviews: relative F1 change vs. $\gamma = 0.5$, computed at the conductance-selected operating point of each γ . “Range” is the spread of Δ_{F1} across the four γ values.

Cluster	$\Delta_{F1}(0.25)$	$\Delta_{F1}(0.75)$	$\Delta_{F1}(1.00)$	Range
Fashion	−7.6%	0.0%	0.0%	7.6%
All Beauty	+183.5%	+7.2%	+237.1%	244.7%
Appliances	−4.5%	−4.5%	−4.5%	4.5%
Gift Cards	−2.6%	+1.4%	+3.2%	5.8%
Magazine	−0.4%	+7.9%	+12.0%	12.4%
Indust./Sci.	degenerate ($F1 < 0.04$ at all γ)			—
Lux. Beauty	−70.6%	−57.3%	+57.6%	128.2%
Pantry	−1.5%	+34.9%	+103.8%	138.7%
Software	3.1%	−1.8%	−5.6%	8.7%

Sensitivity classes. The two tables together induce a natural grouping of clusters by their γ -response:

- **Class A (γ -robust).** Fashion, Appliances, Gift Cards, and Software exhibit relative F1 ranges below 10% across the entire γ grid. The conductance-selected operating point and the resulting clusters are largely insensitive to γ on these instances.

- **Class B (monotone in γ).** Magazine, and Pantry, with both F1 and Φ improving toward larger γ . The structural-commitment factor de-prioritizes boundary vertices weakly attached to the active region, and these clusters benefit from that filtering at every step of the grid.
- **Class C (non-monotone).** All Beauty and Lux. Beauty exhibit the largest F1 ranges in Table 18 (244.7% and 128.2%, respectively) with non-monotone shapes. Both clusters have a local F1 minimum at or near $\gamma = 0.5$. We also observed that the corresponding conductance values are non-monotone, indicating that the F1 swings reflect genuine changes in the conductance landscape rather than seed-sampling noise alone.
- **Class D (degenerate).** Indust./Sci. has $F1(\gamma) < 0.04$ at every γ . Its conductance trend is approximately monotone, but its F1 is too small for relative changes to be meaningful.

Interpretation. The ablation supports the following observations.

First, on the four Class A clusters, γ is effectively a free parameter: the conductance-selected operating point and the resulting F1 are stable within the noise floor across a $4\times$ range of γ . The value $\gamma = 0.5$ is a defensible default for these clusters.

Second, the structural-commitment factor and the conductance-based selection rule do interact. On Class B and Class C clusters, varying γ changes both the score ranking of boundary vertices and the operating point that the selection rule lands on. The two effects compound: the same γ that produces a different score ordering also induces a different conductance landscape over the f-grid, and the rule then selects a different f. This is visible in the non-monotone Class C entries, where small changes in γ produce large changes in both F1 and conductance.

Compute resources. All experiments were run on a single core of a 12-core Apple M4 Pro processor with 24 GB of RAM. No GPU acceleration was used; both TL-HFD and all baselines are implemented in Julia and executed sequentially.